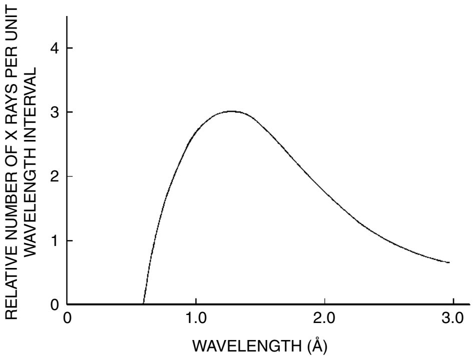
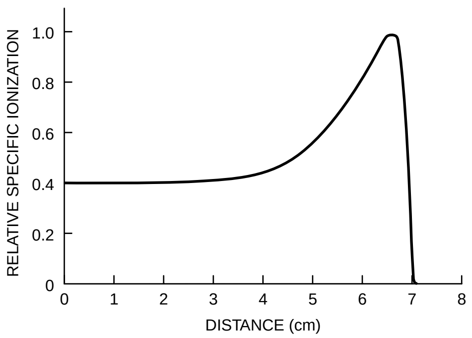
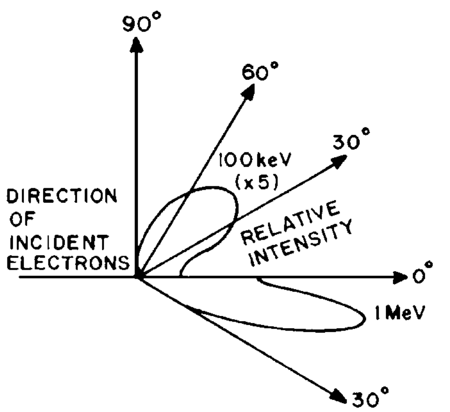


MARGIN FIGURE 4-4
Bremsstrahlung spectrum for a molybdenum target bombarded by electrons accelerated through 20 kV plotted as a function of wavelength in angstroms (Å) $= 10^{-10}$ m. (From Wehr, M., and Richards, J. *Physics of the Atom*. Reading, MA, Addison-Wesley, 1960, p. 159. Used with permission.)



MARGIN FIGURE 4-5
The relative specific ionization of 7.7-MeV α-particles from the decay of ^{214}Po is plotted as a function of the distance traversed in air.



MARGIN FIGURE 4-6
Relative intensity of bremsstrahlung radiated at various angles for electrons with kinetic energies of 100 keV and 1 MeV. (Data from Scherzer, O. *Ann Phys* 1932; **13**:137; and Andrews, H. *Radiation Physics*. Englewood Cliffs, NJ, Prentice-Hall International, 1961.)

INTERACTIONS OF HEAVY, CHARGED PARTICLES

Protons, deuterons, α-particles, and other heavy, charged particles lose kinetic energy rapidly as they penetrate matter. Most of the energy is lost as the particles interact inelastically with electrons of the absorbing medium. The transfer of energy is accomplished by interacting electrical fields, and physical contact is not required between the incident particles and absorber electrons. From Examples 4-1 and 4-2, it is apparent that α-particles produce dense patterns of interaction but have limited range. Deuterons, protons, and other heavy, charged particles also exhibit high specific ionization and a relatively short range. The density of soft tissue (1 g/cm^3) is much greater than the density of air (1.29×10^{-3} g/cm^3). Hence, α-particles of a few MeV or less from radioactive nuclei penetrate soft tissue to depths of only a few microns ($1\ \mu\text{m} = 10^{-6}$m). For example, α-particles from a radioactive source near or on the body penetrate only the most superficial layers of the skin.

The specific ionization (SI) and LET are not constant along the entire path of monoenergetic charged particles traversing a homogeneous medium. The SI of 7.7-MeV α-particles from ^{214}Po is plotted in Margin Figure 4-5 as a function of the distance traversed in air. The increase of SI near the end of the path of the particles reflects the decreased velocity of the α-particles. As the particles slow down, the SI increases because nearby atoms are influenced for a longer period of time. The region of increased SI is termed the Bragg peak. The rapid decrease in SI beyond the peak is due primarily to the capture of electrons by slowly moving α-particles. Captured electrons reduce the charge of the α-particles and decrease their ability to produce ionization.

INDIRECTLY IONIZING RADIATION

Uncharged particles such as neutrons and photons are said to be indirectly ionizing. Neutrons have no widespread application at the present time in medical imaging. They are discussed here briefly to complete the coverage of the "fundamental" particles that make up the atom.

INTERACTIONS OF NEUTRONS

Neutrons may be produced by a number of sources. The distribution of energies available depends on the method by which the "free" neutrons are produced. Slow, intermediate, and fast neutrons (Table 4-1) are present within the core of a nuclear reactor. Neutrons with various kinetic energies are emitted by ^{252}Cf, a nuclide that fissions spontaneously. This nuclide has been encapsulated into needles and used experimentally for implant therapy. Neutron beams are available from cyclotrons and other accelerators in which low-Z nuclei (e.g., ^{3}H or ^{9}Be) are bombarded by positively charged particles (e.g., nuclei of ^{1}H, ^{2}H, ^{3}H) moving at high velocities. Neutrons are released as a product of this bombardment. The energy distribution of neutrons from these devices depends on the target material and on the type and energy of the bombarding particle.

Neutrons are uncharged particles that interact primarily by "billiard ball" or "knock-on" collisions with absorber nuclei. A knock-on collision is elastic if the kinetic energy of the particles is conserved. The collision is inelastic if part of the kinetic energy is used to excite the nucleus. During an elastic knock-on collision, the energy transferred from a neutron to the nucleus is maximum if the mass of the nucleus equals the neutron mass. If the absorbing medium is tissue, then the energy transferred per collision is greatest for collisions of neutrons with nuclei of hydrogen, because the mass of the hydrogen nucleus (i.e., a proton) is close to the mass of a neutron. Most nuclei in tissue are hydrogen, and the cross section for elastic collision is greater for hydrogen than for other constituents of tissue. For these reasons, elastic collisions with hydrogen nuclei account for most of the energy deposited in tissue by neutrons with kinetic energies less than 10 MeV.

For neutrons with kinetic energy greater than 10 MeV, inelastic scattering also contributes to the energy lost in tissue. Most inelastic interactions occur with nuclei other than hydrogen. Energetic charged particles (e.g., protons or α-particles) are often ejected from nuclei excited by inelastic interactions with neutrons.

ATTENUATION OF X AND γ RADIATION

When an x or γ ray impinges upon a material, there are three possible outcomes. The photon may (1) be absorbed (i.e., transfer its energy to atoms of the target material) during one or more interactions; (2) be scattered during one or more interactions; or (3) traverse the material without interaction. If the photon is absorbed or scattered, it is said to have been attenuated. If 1000 photons impinge on a slab of material, for example, and 200 are scattered and 100 are absorbed, then 300 photons have been attenuated from the beam, and 700 photons have been transmitted without interaction.

Attenuation processes can be complicated. Partial absorption may occur in which only part of the photon's energy is retained in the absorber. Scattering through small angles may not remove photons from the beam, especially if the beam is rather broad.

TABLE 4-1 Classification of Neutrons According to Kinetic Energy

Type	*Energy Range*
Slow	0–0.1 keV
Intermediate	0.1–20 keV
Fast	20 keV–10 Mev
High-energy	>10 MeV

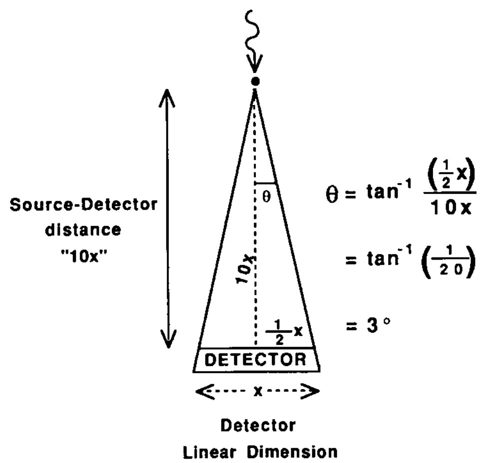
Attenuation of a Beam of X or γ Rays

The number of photons attenuated in a medium depends on the number transversing the medium. If all the photons posses the same energy (i.e., the beam is monoenergetic) and if the photons are attenuated under conditions of good geometry (i.e., the beam is narrow and the transmitted beam contains no scattered photons), then the number I of photons penetrating a thin slab of matter of thickness x is

$$I = I_0 e^{-\mu x} \tag{4-6}$$

where μ is the attenuation coefficient of the medium for the photons and I_0 represents the number of photons in the beam before the thin slab of matter is placed into position. The number I_{at} of photons attenuated (absorbed or scattered) from the beam is

$$\begin{aligned} I_{\text{at}} &= I_0 - I \\ &= I_0 - I_0 e^{-\mu x} \\ &= I_0(1 - e^{-\mu x}) \end{aligned} \tag{4-7}$$



MARGIN FIGURE 4-7
Illustration of "small-angle" scatter. Scatter of greater than 3 degrees generally is accepted as attenuation (removal) from a narrow beam. A generally accepted definition of a broad beam is a beam that is broader than one-tenth the distance to a point source of radiation.

The exponent of e must possess no units. Therefore, the units for μ are 1/cm if the thickness x is expressed in centimeters, 1/in. if x is expressed in inches, and so on. An attenuation coefficient with units of 1/length is called a linear attenuation coefficient. Derivation of Eqs. (4-6), (4-7), and other exponential expressions is given in Appendix I.

The mean path length is the average distance traveled by x or γ rays before interaction in a particular medium. The mean path length is sometimes termed the mean free path or relaxation length and equals $1/\mu$, where μ is the total linear attenuation coefficient. Occasionally the thickness of an attenuating medium may be expressed in multiples of the mean free path of photons of a particular energy in the medium.

The probability is $e^{-\mu x}$ that a photon traverses a slab of thickness x without interacting. This probability is the product of probabilities that the photon does not interact by any of five interaction processes:

$$e^{-\mu x} = (e^{-\omega x})\,(e^{-\tau x})\,(e^{-\sigma x})\,(e^{-\kappa x})\,(e^{-\pi x}) = e^{-(\omega+\tau+\sigma+\kappa+\pi)x}$$

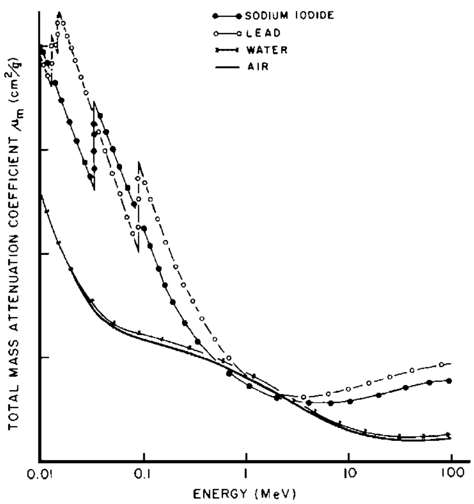
The coefficients ω, τ, σ, κ, and π represent attenuation by the processes of coherent scattering (ω), photoelectric absorption (τ), Compton scattering (σ), pair production (κ), and photodisintegration (π), as described later. The total linear attenuation coefficient may be written

$$\mu = \omega + \tau + \sigma + \kappa + \pi \qquad \textbf{(4-8)}$$

In diagnostic radiography, coherent scattering, photodisintegration, and pair production are usually negligible, and μ is written

$$\mu = \tau + \sigma \qquad \textbf{(4-9)}$$

In general, attenuation coefficients vary with the energy of the x or γ rays and with the atomic number of the absorber. Linear attenuation coefficients also depend on the density of the absorber. Mass attenuation coefficients, obtained by dividing linear attenuation coefficients by the density ρ of the attenuating medium, do not vary with the density of the medium:



MARGIN FIGURE 4-8
Mass attenuation coefficients for selected materials as a function of photon energy.

$$\mu_m = \frac{\mu}{\rho}, \qquad \omega_m = \frac{\omega}{\rho}, \qquad \tau_m = \frac{\tau}{\rho}, \qquad \sigma_m = \frac{\sigma}{\rho}, \qquad \kappa_m = \frac{\kappa}{\rho}, \qquad \pi_m = \frac{\pi}{\rho}$$

Mass attenuation coefficients usually have units of square meters per kilogram, although sometimes square centimeters per gram and square centimeters per milligram are used. Total mass attenuation coefficients for air, water, sodium iodide, and lead are plotted in Margin Figure 4-8 as a function of the energy of incident photons.

When mass attenuation coefficients are used, thicknesses x_m are expressed in units such as kilograms per square meter, grams per square centimeter, or milligrams per square centimeter. A unit of thickness such as kilograms per square meter may seem unusual but can be understood by recognizing that this unit describes the mass of a 1-m^2 cross section of an absorber. That is, the amount of material traversed by a beam is indirectly measured by the mass per unit area of the slab. The thickness of attenuating medium may also be expressed as the number of atoms or electrons per unit area. Symbols x_a and x_e denote thicknesses in units of atoms per square meter and electrons per square meter. These thicknesses may be computed from the linear thickness x by

$$\begin{aligned} X_a \frac{\text{atoms}}{\text{m}^2} &= \frac{x(\text{m})\rho(kg/\text{m}^3)N_a(\text{atoms/gram-atomic mass})}{M(\text{kg/gram-atomic mass})} \\ &= \frac{x\rho N_a}{M} \end{aligned} \qquad \textbf{(4-10)}$$

$$\begin{aligned} x_e \frac{\text{electrons}}{\text{m}^2} &= \left[x_a\left(\frac{\text{atoms}}{\text{m}^2}\right)\right]\left[Z\left(\frac{\text{electrons}}{\text{atom}}\right)\right] \\ &= x_a Z \end{aligned} \qquad \textbf{(4-11)}$$

In these equations, M is the gram-atomic mass of the attenuating medium, Z is the atomic number of the medium, and N_a is Avogadro's number (6.02×10^{23}), the number of atoms per gram-atomic mass. Total atomic and electronic attenuation coefficients μ_a and μ_e, to be used with thicknesses x_a and x_e, may be computed from the linear attenuation coefficient μ by

$$\begin{aligned}\mu_a\left(\frac{\text{m}^2}{\text{atom}}\right) &= \frac{\mu(\text{m}^{-1})M(\text{kg/gram-atomic mass})}{\rho(kg/\text{m}^3)N_a(\text{atoms/gram-atomic mass})} \\ &= \frac{\mu M}{\rho N_a}\end{aligned} \tag{4-12}$$

$$\begin{aligned}\mu_e\left(\frac{\text{m}^2}{\text{electron}}\right) &= \frac{\mu_a(\text{m}^2/\text{atom})}{Z(\text{electrons/atom})} \\ &= \frac{\mu_a}{Z}\end{aligned} \tag{4-13}$$

The number I of photons penetrating a thin slab of matter may be computed with any of the following expressions:

$$I = I_0 e^{-\mu x} \qquad I = I_0 e^{-\mu_a x_a}$$
$$I = I_0 e^{-\mu_m x_m} \qquad I = I_0 e^{-\mu_e x_e}$$

Example 4-4

A narrow beam containing 2000 monoenergetic photons is reduced to 1000 photons by a slab of copper 10^{-2} m thick. What is the total linear attenuation coefficient of the copper slab for these photons?

$$\begin{aligned}I &= I_0 e^{-\mu x} \\ I/I_0 &= e^{-\mu x} \\ I_0/I &= e^{\mu x} \\ \text{In } I_0/I &= \mu x \\ \text{In}\frac{2000 \text{ photons}}{1000 \text{ photons}} &= \mu(10^{-2}\text{ m}) \\ \text{In } 2 &= \mu(10^{-2}\text{ m}) \\ \mu &= \frac{\text{In } 2}{10^{-2}\text{ m}} \\ \mu &= \frac{0.693}{10^{-2}\text{ m}} \\ \mu &= 69.3\text{ m}^{-1}\end{aligned}$$

The thickness of a slab of matter required to reduce the intensity (or exposure rate—see Chapter 6) of an x- or γ-ray beam to half is the half-value layer (HVL) or half-value thickness (HVT) for the beam. The HVL describes the "quality" or penetrating ability of the beam. The HVL in Example 4-4 is 10^{-2} m of copper. The HVL of a monoenergetic beam of x or γ rays in any medium is

$$\text{HVL} = \frac{\ln 2}{\mu} \tag{4-14}$$

where $\ln 2 = 0.693$ and μ is the total linear attenuation coefficient of the medium for photons in the beam. The measurement of HVLs for monoenergetic and polyenergetic beams is discussed in Chapter 6.

Example 4-5

a. What are the total mass (μ_m), atomic (μ_a), and electronic (μ_e) attenuation coefficients of the copper slab described in Example 4-4? Copper has a density of 8.9×10^3 kg/m^3, a gram-atomic mass of 63.6, and an atomic number of 29.

$$\begin{aligned}\mu_m &= \mu/\rho \\ &= \frac{69.3 \text{ m}^{-1}}{8.9 \times 10^3 \text{ kg/m}^3} \\ &= 0.0078 \text{ m}^2\text{/kg}\end{aligned}$$

$$\begin{aligned}\mu_a &= \frac{\mu M}{\rho N} \\ &= \frac{(69.3 \text{ m}^{-1})(63.6 \text{ g/gram-atomic mass})(10^{-3} \text{ kg/g})}{(8.9 \times 10^3 \text{ kg/m}^3)(6.02 \times 10^{23} \text{ atoms/gram-atomic mass})} \\ &= 8.2 \times 10^{-28} \text{m}^2\text{/atom}\end{aligned} \tag{4-15}$$

$$\begin{aligned}\mu_e &= \frac{\mu_a}{Z} \\ &= \frac{8.2 \times 10^{-28} \text{m}^2\text{/atom}}{29 \text{ electrons/atom}} \\ &= 2.8 \times 10^{-29} \text{m}^2\text{/electron}\end{aligned} \tag{4-16}$$

b. To the 0.01-m copper slab, 0.02 m of copper is added. How many photons remain in the beam emerging from the slab?

$$\begin{aligned}I &= I_0 e^{-\mu x} \\ &= (2000 \text{ photons})e^{-(69.3 \text{ m}^{-1})(0.03 \text{ m})} \\ &= (2000 \text{ photons})\text{e}^{-2.079} \\ &= (2000 \text{ photons})(0.125) \\ &= 250 \text{ photons}\end{aligned} \tag{4-17}$$

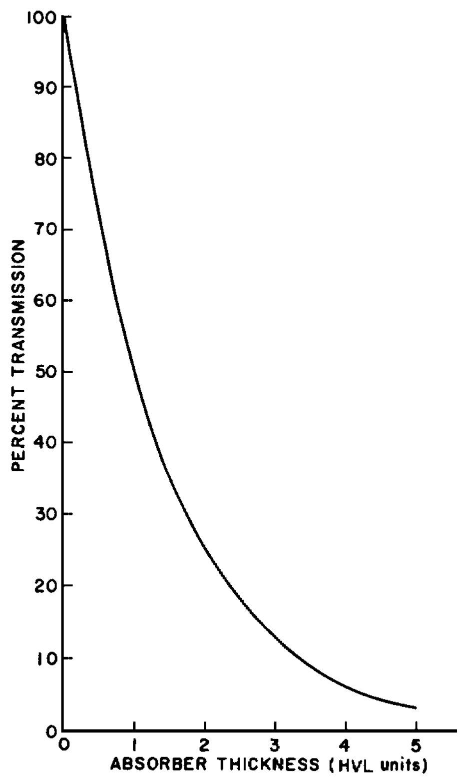
A thickness of 0.01 m of copper reduces the number of photons to half. Because the beam is narrow and monoenergetic, adding two more of the same thicknesses of copper reduces the number to one eighth. Only one eighth of the original number of photons remains after the beam has traversed three thicknesses of copper.

c. What is the thickness x_e in electrons per square meter for the 0.03-m slab?

$$\begin{aligned}x_e &= x_a Z \\ &= \frac{x N_a \rho Z}{M} \\ &= \frac{(0.03 \text{ m})(8.9 \times 10^3 \, kg\text{/m}^3)(6.02 \times 10^{23} \text{ atoms/gram-atomic mass})(29 \text{ electrons/atom})}{(63.3 \text{ g/gram-atomic mass})(10^{-3} \text{ kg/g})} \\ &= 7.3 \times 10^{28} \text{ electrons/m}^2\end{aligned}$$

d. Repeat the calculation in Part b, but use the electronic attenuation coefficient.

$$\begin{aligned}I &= I_0 e^{-\mu_e x_e} \\ &= (2000 \text{ photons})e^{-(2.8\times10^{-29} \text{ m}^2/\text{electron})(7.3\times10^{28} \text{ electrons/m}^2)} \\ &= (2000 \text{ photons})e^{-2.079}\end{aligned}$$



MARGIN FIGURE 4-9
Percent transmission of a narrow beam of monoenergetic photons as a function of the thickness of an attenuating slab in units of half-value layer (HVL). Absorption conditions satisfy requirements for "good geometry."

$$= (2000 \text{ photons})(0.125)$$
$$= 250 \text{ photons}$$

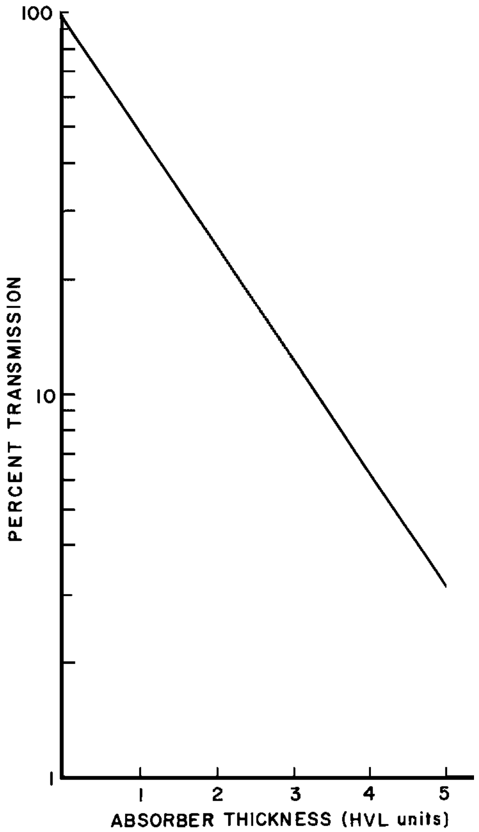
A narrow beam of monoenergetic photons is attenuated exponentially. Exponential attenuation is described by Eq. (4-6). From Example 4-4 we have the equivalent relationship:

$$\ln \frac{I}{I_0} = -\mu x$$

The logarithm of the number I of photons decreases linearly with increasing thickness of the attenuating slab. Hence a straight line is obtained when the logarithm of the number of x or γ rays is plotted as a function of thickness. When the log of the dependent (y-axis) variable is plotted as a function of the linear independent (x-axis) variable, the graph is termed a semilog plot. It should be emphasized that a semilogarithmic plot of the number of photons versus the thickness of the attenuating slab yields a straight line only if all photons possess the same energy and the conditions for attenuation fulfill requirements for narrow-beam geometry.

If broad-beam geometry were used, the falloff of the number of photons as a function of slab thickness would be less rapid. In broad-beam geometry, some scattered photons continue to strike the detector and are therefore not considered to have been "attenuated." A broad-beam measurement is relevant when considering the design of wall shielding for radiation safety near x-ray equipment. Humans on the opposite side of barriers would encounter a broad beam. Narrow-beam geometry is more relevant to situations such as experimental studies of the properties of materials.

When an x-ray beam is polyenergetic (as is the beam emitted from an x-ray tube), a plot of the number of photons remaining in the beam as a function of the thickness of the attenuating material it has traversed does not yield a straight line on a semilogarithmic plot. This is because the attenuation of a polyenergetic beam cannot be represented by a single simple exponential equation with a single attenuation coefficient. It is represented by a weighted average of exponential equations for each of the different photon energies in the beam.



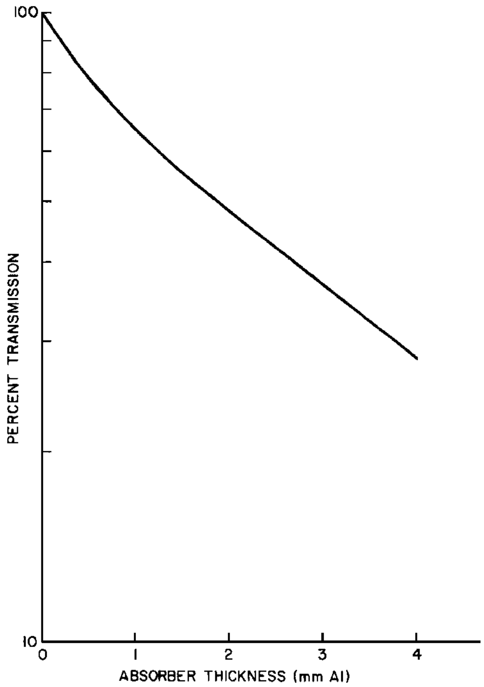
MARGIN FIGURE 4-10
Semilogarithmic plot of data in Margin Figure 4-9.

Filtration

In a polyenergetic beam, lower-energy photons are more likely to be attenuated and higher energy photons are more likely to pass through without interaction. Therefore, after passing through some attenuating material, the distribution of photon energies contained within the emergent beam is different than that of the beam that entered. The emergent beam, although containing fewer total photons, actually has a higher average photon energy. This effect is known as "filtration." A beam that has undergone filtration is said to be "harder" because it has a higher average energy and therefore more penetrating power.

To remove low-energy photons and increase the penetrating ability of an x-ray beam from a diagnostic or therapy x-ray unit, filters of aluminum, copper, tin, or lead may be placed in the beam. For most diagnostic x-ray units, the added filters are usually 1 to 3 mm of aluminum. Because the energy distribution changes as a polyenergetic x-ray beam penetrates an attenuating medium, no single value for the attenuation coefficient may be used in Eq. (4-6) to compute the attenuation of the x-ray beam. However, from the measured half-value layer (HVL) an effective attenuation coefficient may be computed (see Example 4-6):

$$\mu_{\text{eff}} = \frac{\ln 2}{\text{HVL}}$$



MARGIN FIGURE 4-11
Semilogarithmic plot of the number of x rays in a polyenergetic beam as a function of thickness of an attenuating medium. The penetrating ability (HVL) of the beam increases continuously with thickness because lower-energy photons are selectively removed from the beam. The straight-line relationship illustrated in Margin Figure 4-10 for a narrow beam of monoenergetic photons is not achieved for a polyenergetic x-ray beam.

The effective energy of an x-ray beam is the energy of monoenergetic photons that have an attenuation coefficient in a particular medium equal to the effective attenuation coefficient for the x-ray beam in the same medium.

Example 4-6

An x-ray beam produced at 200 kVp has an HVL of 1.5 mm Cu. The density of copper is 8900 kg/m^3.

a. What are the effective linear and mass attenuation coefficients?

$$\begin{aligned}
\mu_{\text{eff}} &= \frac{\ln 2}{1.5 \text{ mm Cu}} \\
&= 0.46 \ (\text{mm Cu})^{-1} \\
(\mu_m)_{\text{eff}} &= \frac{\mu_{\text{eff}}}{\rho} \\
&= \frac{0.46\,(\text{mm Cu})^{-1}(10^3 \text{ mm/m})}{8.9 \times 10^3 \text{ kg/m}^3} \\
&= 0.052 \text{ m}^2/\text{kg}
\end{aligned}$$

b. What is the average effective energy of the beam?

Monoenergetic photons of 96 keV possess a total mass attenuation coefficient of 0.052 m^2/kg in copper. Consequently, the average effective energy of the x-ray beam is 96 keV.

Energy Absorption and Energy Transfer

The attenuation coefficient μ (or μ_a, μ_e, μ_m) refers to total attenuation (i.e., absorption plus scatter). Sometimes it is necessary to determine the energy truly absorbed in a material and not simply scattered from it. To express the energy absorbed, the energy absorption coefficient μ_{en} is used, where μ_{en} is given by the expression

$$\mu_{\text{en}} = \mu \frac{E_a}{h\nu} \tag{4-18}$$

In this expression, μ is the attenuation coefficient, E_a is the average energy absorbed in the material per photon interaction, and $h\nu$ is the photon energy. Thus, the energy absorption coefficient is equal to the attenuation coefficient times the fraction of energy truly absorbed. Energy absorption coefficients may also be expressed as mass energy absorption coefficients $(\mu_{\text{en}})_m$, atomic energy absorption coefficients $(\mu_{\text{en}})_a$, or electronic energy absorption coefficients $(\mu_{\text{en}})_e$ by dividing the energy absorption coefficient μ_{en} by the physical density, the number of atoms per cubic meter, or the number of electrons per cubic meter, respectively.

Example 4-7

The attenuation coefficient for 1-MeV photons in water is 7.1 m^{-1}. If the energy absorption coefficient for 1-MeV photons in water is 3.1 m^{-1}, find the average energy absorbed in water per photon interaction.

By rearranging Eq. (4-15), we obtain

$$\begin{aligned}
E_a &= \frac{\mu_{\text{en}}}{\mu}(h\nu) \\
&= \frac{3.1 \text{ m}^{-1}}{7.1 \text{ m}^{-1}}(1 \text{ MeV})
\end{aligned}$$

$$= 0.44 \text{ MeV}$$
$$= 440 \text{ keV}$$

Example 4-8

An x-ray tube emits 10^{12} photons per second in a highly collimated beam that strikes a 0.l-mm-thick radiographic screen. For purposes of this example, the beam is assumed to consist entirely of 40-keV photons. The attenuation coefficient of the screen is 23 m^{-1}, and the mass energy absorption coefficient of the screen is 5 m^{-1} for 40-keV photons. Find the total energy in keV absorbed by the screen during a 0.5-sec exposure.

The number of photons incident upon the screen is

$$(10^{12} \text{ photons/sec})(0.5 \text{ sec}) = 5 \times 10^{11} \text{ photons}$$

The number of interactions that take place in the screen is [from Eq. (4-7)]

$$\begin{aligned} I_{\text{at}} &= I_0(1 - e^{-\mu x}) \\ &= 5 \times 10^{11}\left[1 -^{-(23 \text{ m}^{-1})(10^{-3} \text{ m/mm})}\right] \\ &= 1.2 \times 10^9 \text{ interactions} \end{aligned}$$

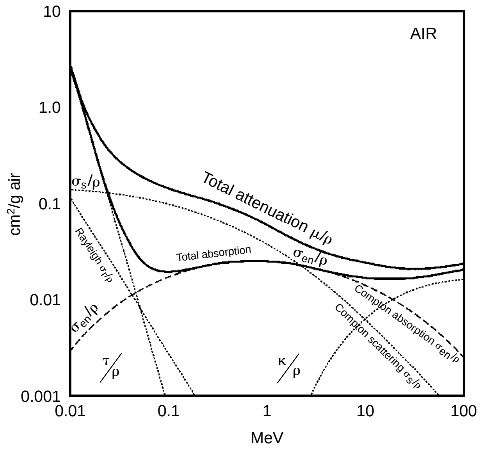
The average energy absorbed per interaction is [from Eq. (4-15)]

$$\begin{aligned} E_a &= \frac{\mu_{\text{en}}}{\mu}(h\nu) \\ &= \frac{5 \text{ m}^{-1}}{23 \text{ m}^{-1}}(40 \text{ keV}) \\ &= 8.7 \text{ keV} \end{aligned}$$

The total energy absorbed during the 0.5-sec exposure is then

$$(1.2 \times 10^9)(8.7 \text{ keV}) = 1.3 \times 10^{10} \text{ keV}$$

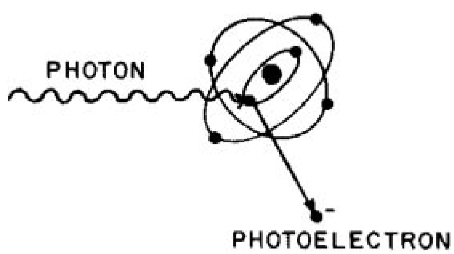
Energy absorption coefficients and attenuation coefficients for air are plotted in Margin Figure 4-12 as a function of photon energy.



MARGIN FIGURE 4-12
Mass attenuation coefficients for photons in air. The curve marked “total absorption” is $(\mu/\rho) = (\sigma/\rho) + (\tau/\rho) + (\kappa/\rho)$, where σ, τ, and κ are the corresponding linear coefficients for Compton absorption, photoelectric absorption, and pair production. When the Compton mass scattering coefficient σ_s and the Compton absorption coefficient τ_a/ρ are added, the total Compton mass attenuation coefficient σ/ρ is obtained.

Coherent Scattering: Ionizing Radiation

Photons are deflected or scattered with negligible loss of energy by the process of coherent or Rayleigh scattering. Coherent scattering is sometimes referred to as classical scattering because the interaction may be completely described by methods of classical physics. The classical description assumes that a photon interacts with electrons of an atom as a group rather than with a single electron within the atom. Usually the photon is scattered in approximately the same direction as the incident photon. Although photons with energies up to 150 to 200 keV may scatter coherently in media with high atomic number, this interaction is important in tissue only for low-energy photons. The importance of coherent scattering is further reduced because little energy is deposited in the attenuating medium. However, coherent scatter sometimes reduces the resolution of scans obtained with low-energy, γ-emitting nuclides (e.g., ^{125}I) used in nuclear medicine.

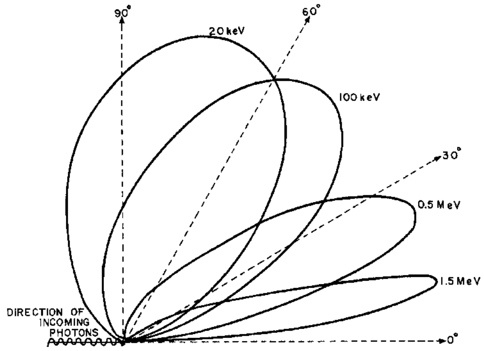


MARGIN FIGURE 4-13
Photoelectric absorption of a photon with energy $h\nu$. The photon disappears and is replaced by an electron ejected from the atom with kinetic energy $E_k = h\nu - E_b$, where E_b is the binding energy of the electron. Characteristic radiation and Auger electrons are emitted as electrons cascade to replace the ejected photoelectron.

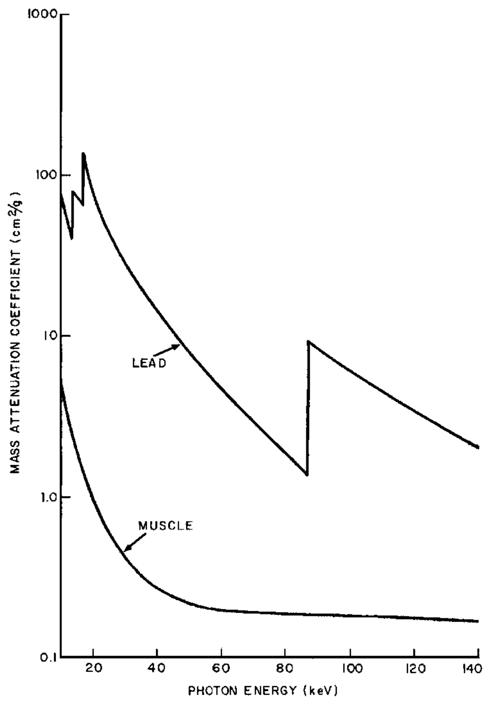
Photoelectric Absorption

During a photoelectric interaction, the total energy of an x or γ ray is transferred to an inner electron of an atom. The electron is ejected from the atom with kinetic energy E_κ, where E_κ equals the photon energy $h\nu$ minus the binding energy E_B required

Albert Einstein (1879–1955) was best known to the general public for his theories of relativity. However, he also developed a theoretical explanation of the photoelectric effect—that the energy of photons in a light beam do not add together. It is the energy of each photon that determines whether the beam is capable of removing inner shell electrons from an atom. His Nobel Prize of 1921 mentioned only his explanation of the photoelectric effect and did not cite relativity because that theory was somewhat controversial at that time.



MARGIN FIGURE 4-14
Electrons are ejected approximately at a right angle as low-energy photons interact photoelectrically. As the energy of the photons increases, the angle decreases between incident photons and ejected electrons.



MARGIN FIGURE 4-15
Photoelectric mass attenuation coefficients of lead and soft tissue as a function of photon energy. K- and L-absorption edges are depicted for lead.

to remove the electron from the atom:

$$E_\kappa = h\nu - E_B \tag{4-19}$$

The ejected electron is called a photoelectron.

Example 4-9

What is the kinetic energy of a photoelectron ejected from the K shell of lead (E_B = 88 keV) by photoelectric absorption of a photon of 100 keV?

$$\begin{aligned} E_\kappa &= h\nu - E_B \\ &= 100 \text{ keV} - 88 \text{ keV} \\ &= 12 \text{ keV} \end{aligned}$$

The average binding energy is only 0.5 keV for K electrons in soft tissue. Consequently, a photoelectron ejected from the K shell of an atom in tissue possesses a kinetic energy about 0.5 keV less than the energy of the incident photon.

Photoelectrons resulting from the interaction of low-energy photons are released approximately at a right angle to the motion of the incident photons. As the energy of the photons increases, the average angle decreases between incident photons and released photoelectrons.

An electron ejected from an inner shell leaves a vacancy or hole that is filled immediately by an electron from an energy level farther from the nucleus. Only rarely is a hole filled by an electron from outside the atom. Instead, electrons usually cascade from higher to lower energy levels and produce a number of characteristic photons and Auger electrons with energies that, when added together, equal the binding energy of the ejected photoelectron. Characteristic photons and Auger electrons released during photoelectric interactions in tissue possess an energy less than 0.5 keV. These lower-energy photons and electrons are absorbed rapidly in surrounding tissue.

The probability of photoelectric interaction decreases rapidly as the photon energy increases. In general, the mass attenuation coefficient τ_m for photoelectric absorption varies roughly as $1/(h\nu)^3$, where $h\nu$ is the photon energy. In Margin Figure 4-15, the photoelectric mass attenuation coefficients τ_m of muscle and lead are plotted as a function of the energy of incident photons. Discontinuities in the curve for lead are termed *absorption edges* and occur at photon energies equal to the binding energies of electrons in inner electron shells. Photons with energy less than the binding energy of K-shell electrons interact photoelectrically only with electrons in the L shell or shells farther from the nucleus. Photons with energy equal to or greater than the binding energy of K-shell electrons interact predominantly with K-shell electrons. Similarly, photons with energy less than the binding energy of L-shell electrons interact only with electrons in M and more distant shells. That is, most photons interact photoelectrically with electrons that have a binding energy nearest to but less than the energy of the photons. Hence, the photoelectric attenuation coefficient increases abruptly at photon energies equal to the binding energies of electrons in different shells. Absorption edges for photoelectric attenuation in soft tissue occur at photon energies that are too low to be shown in Margin Figure 4-15. Iodine and barium exhibit K-absorption edges at energies of 33 and 37 keV. Compounds containing these elements are routinely used as contrast agents in diagnostic radiology.

At all photon energies depicted in Margin Figure 4-15, the photoelectric attenuation coefficient for lead ($Z = 82$) is greater than that for soft tissue ($Z_{\text{eff}} = 7.4$) (Z_{eff} represents the effective atomic number of a mixture of elements.) In general, the photoelectric mass attenuation coefficient varies with Z^3. For example, the number of 15-keV photons absorbed primarily by photoelectric interaction in bone ($Z_{\text{eff}} = 11.6$) is approximately four times greater than the number of 15-keV photons absorbed in an equal mass of soft tissue because $(11.6/7.4)^3 = 3.8$. Selective attenuation of photons in media with different atomic numbers and different physical densities is the principal reason for the usefulness of low-energy x rays for producing images in diagnostic radiology.

Effective atomic number (Z_{eff}): A pure element, a material composed of only one type of atom, is characterized by its atomic number (the number of protons in its nucleus). In a composite material, the *effective* atomic number is a weighted average of the atomic numbers of the different elements of which it is composed. The average is weighted according to the relative number of each type of atom.

Compton (Incoherent) Scattering

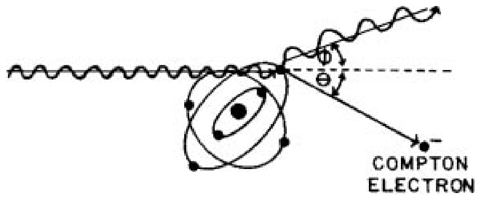
X and γ rays with energy between 30 keV and 30 MeV interact in soft tissue predominantly by Compton scattering. During a Compton interaction, part of the energy of an incident photon is transferred to a loosely bound or "free" electron within the attenuating medium. The kinetic energy of the recoil (Compton) electron equals the energy lost by the photon, with the assumption that the binding energy of the electron is negligible. Although the photon may be scattered at any angle ϕ with respect to its original direction, the Compton electron is confined to an angle θ, which is 90 degrees or less with respect to the motion of the incident photon. Both θ and ϕ decrease with increasing energy of the incident photon (Margin Figure 4-17).

During a Compton interaction, the change in wavelength [$\Delta\lambda$ in nanometers (nm), 10^{-9} meters] of the x or γ ray is

$$\Delta\lambda = 0.00243\,(1 - \cos\phi) \tag{4-20}$$

where ϕ is the scattering angle of the photon. The wavelength of the scattered photon is

$$\lambda' = \lambda + \Delta\lambda \tag{4-21}$$

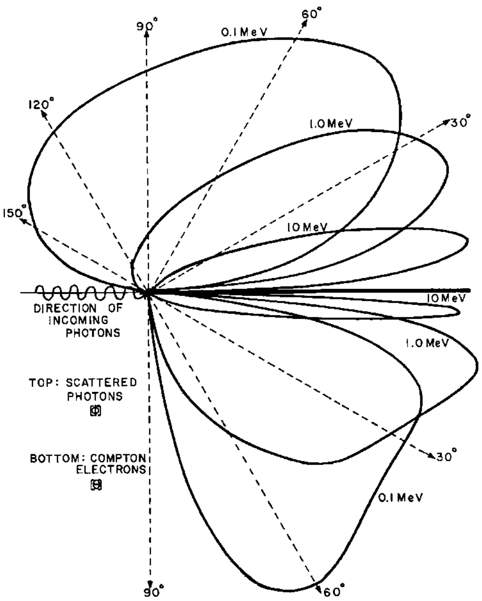


MARGIN FIGURE 4-16
Compton scattering of an incident photon, with the photon scattered at an angle ϕ; the Compton electron is ejected at an angle θ with respect to the direction of the incident photon.

where λ is the wavelength of the incident photon. The energies $h\nu$ and $h\nu'$ of the incident and scattered photons are

$$\begin{aligned} h\nu\ (\text{keV}) &= \frac{hc}{\lambda} \\ &= \frac{(6.62\times10^{-34}\ \text{J-sec})(3\times10^{8}\ \text{m/sec})}{(\lambda)(10^{-9}\ \text{m/nm})(1.6\times10^{-19}\ \text{J/eV})(10^{3}\ \text{eV/keV})} \\ &= \frac{1.24}{\lambda} \end{aligned} \tag{4-22}$$

with λ expressed in nanometers.



MARGIN FIGURE 4-17
Electron scattering angle θ and photon scattering angle ϕ as a function of the energy of incident photons. Both θ and ϕ decrease as the energy of incident photons increases.

Example 4-10

A 210-keV photon is scattered at an angle of 80 degrees during a Compton interaction. What are the energies of the scattered photon and the Compton electron? The wavelength λ of the incident photon is

$$\begin{aligned} \lambda &= \frac{1.24}{h\nu} \\ &= \frac{1.24}{210\ \text{keV}} \\ &= 0.0059\ \text{nm} \end{aligned} \tag{4-23}$$

The change in wavelength $\Delta\lambda$ is

$$\begin{aligned} \Delta\lambda &= 0.00243\,(1 - \cos\phi) \\ &= 0.00243\,(1 - \cos(80\ \text{degrees})) \\ &= 0.00243\,(1 - 0.174) \\ &= 0.0020\ \text{nm} \end{aligned} \tag{4-24}$$

The wavelength of the scattered photon is

$$\begin{aligned}\lambda' &= \lambda + \Delta\lambda \\ &= (0.0059 + 0.0020)\ \text{nm} \\ &= 0.0079\ \text{nm}\end{aligned} \tag{4-25}$$

The energy of the scattered photon is

$$\begin{aligned}h\nu' &= \frac{1.24}{\lambda} \\ &= \frac{1.24}{0.0079\ \text{nm}} \\ &= 160\ \text{keV}\end{aligned} \tag{4-26}$$

The energy of the Compton electron is

$$\begin{aligned}E_k &= h\nu - h\nu' \\ &= (210 - 160)\ \text{keV} \\ &= 50\ \text{keV}\end{aligned}$$

Example 4-11

A 20-keV photon is scattered by a Compton interaction. What is the maximum energy transferred to the recoil electron?

The energy transferred to the electron is greatest when the change in wavelength of the photon is maximum; $\Delta\lambda$ is maximum when $\phi = 180$ degrees.

$$\begin{aligned}\Delta\lambda_{\text{max}} &= 0.00243\ [1 - \cos(180)] \\ &= 0.00243\ [1 - (-1)] \\ &= 0.00486\ \text{nm} \\ &= 0.005\ \text{nm}\end{aligned}$$

The wavelength λ of a 20-keV photon is

$$\begin{aligned}\lambda &= \frac{1.24}{h\nu} \\ &= \frac{1.24}{20\ \text{keV}} \\ &= 0.062\ \text{nm}\end{aligned} \tag{4-27}$$

The wavelength λ' of the photon scattered at 180 degrees is

$$\begin{aligned}\lambda' &= \lambda + \Delta\lambda \\ &= (0.062 + 0.005)\ \text{nm} \\ &= 0.067\ \text{nm}\end{aligned} \tag{4-28}$$

The energy $h\nu'$ of the scattered photon is

$$\begin{aligned}h\nu' &= \frac{1.24}{\lambda'} \\ &= \frac{1.24}{0.067\ \text{nm}} \\ &= 18.6\ \text{keV}\end{aligned} \tag{4-29}$$

The energy E_k of the Compton electron is

$$\begin{aligned} E_k &= h\nu - h\nu' \\ &= (20.0 - 18.6)\text{ keV} \\ &= 1.4\text{ keV} \end{aligned}$$

When a low-energy photon undergoes a Compton interaction, most of the energy of the incident photon is retained by the scattered photon. Only a small fraction of the energy is transferred to the electron.

Example 4-12

A 2-MeV photon is scattered by a Compton interaction. What is the maximum energy transferred to the recoil electron? The wavelength λ of a 2-MeV photon is

$$\begin{aligned} \lambda &= \frac{1.24}{h\nu} \\ &= \frac{1.24}{2000\text{ keV}} \\ &= 0.00062\text{ nm} \end{aligned} \tag{4-30}$$

The change in wavelength of a photon scattered at 180 degrees is 0.00486 nm (see Example 4-11). Hence, the wavelength λ' of the photon scattered at 180 degrees is

$$\begin{aligned} \lambda' &= \lambda + \Delta\lambda \\ &= (0.00062 + 0.00486)\text{ nm} \\ &= 0.00548\text{ nm} \end{aligned} \tag{4-31}$$

The energy $h\nu'$ of the scattered photon is

$$\begin{aligned} h\nu' &= \frac{1.24}{\lambda'} \\ &= \frac{1.24}{0.00548\text{ nm}} \\ &= 226\text{ keV} \end{aligned} \tag{4-32}$$

The energy E_k of the Compton electron is

$$\begin{aligned} E_k &= h\nu - h\nu' \\ &= (2000 - 226)\text{ keV} \\ &= 1774\text{ keV} \end{aligned}$$

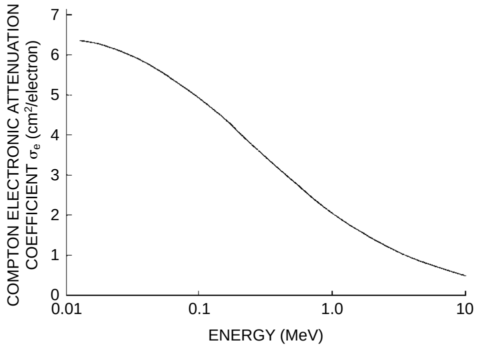
When a high-energy photon is scattered by the Compton process, most of the energy is transferred to the Compton electron. Only a small fraction of the energy of the incident photon is retained by the scattered photon.

Example 4-13

Show that, irrespective of the energy of the incident photon, the maximum energy is 255 keV for a photon scattered at 180 degrees and 511 keV for a photon scattered at 90 degrees.

The wavelength λ' of a scattered photon is

$$\lambda' = \lambda + \Delta\lambda \tag{4-33}$$



MARGIN FIGURE 4-18
Compton electronic attenuation coefficient as a function of photon energy.

MARGIN FIGURE 4-19
Radiographs taken at 70 kVp, 250 kVp, and 1.25 MeV (^{60}Co). These films illustrate the loss of radiographic contrast as the energy of the incident photons increases.

For photons of very high energy, λ is very small and may be neglected relative to $\Delta\lambda$.

For a photon scattered at 180 degrees

$$\begin{aligned}\lambda' \cong \Delta\lambda &= 0.00243\ [1 - \cos(180)]\\ &= 0.00243\ [1 - (-1)]\\ &= 0.00486\text{ nm}\end{aligned}$$

$$\begin{aligned}h\nu' &= \frac{1.24}{\lambda'}\\ &= \frac{1.24}{0.00486\text{ nm}}\\ &= 255\text{ keV}\end{aligned} \tag{4-34}$$

For photons scattered at 90 degrees

$$\begin{aligned}\lambda' \cong \Delta\lambda &= 0.00243\ [1 - \cos(90)]\\ &= 0.00243\ [1 - 0]\\ &= 0.00243\text{ nm}\end{aligned}$$

$$\begin{aligned}h\nu' &= \frac{1.24}{\lambda'}\\ &= \frac{1.24}{0.00243\text{ nm}}\\ &= 511\text{ keV}\end{aligned} \tag{4-35}$$

The Compton electronic attenuation coefficient σ_e is plotted in Margin Figure 4-18 as a function of the energy of incident photons. The coefficient decreases gradually with increasing photon energy. The Compton mass attenuation coefficient varies directly with the electron density (electrons per kilogram) of the absorbing medium because Compton interactions occur primarily with loosely bound electrons. A medium with more unbound electrons will attenuate more photons by Compton scattering than will a medium with fewer electrons.

The Compton mass attenuation coefficient is nearly independent of the atomic number of the attenuating medium. For this reason, radiographs exhibit very poor contrast when exposed to high-energy photons. When most of the photons in a beam of x or γ rays interact by Compton scattering, little selective attenuation occurs in materials with different atomic number. The image in a radiograph obtained by exposing a patient to high-energy photons is not the result of differences in atomic number between different regions of the patient. Instead, the image reflects differences in physical density (kilograms per cubic meter) between the different regions (e.g., bone and soft tissue). The loss of radiographic contrast with increasing energy of incident photons is depicted and discussed more completely in Chapter 7.

Pair Production

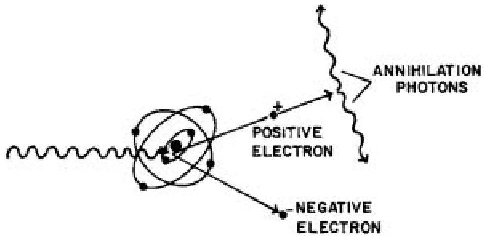
An x or γ ray may interact by pair production while near a nucleus in an attenuating medium. A pair of electrons, one negative and one positive, appears in place of the photon. Because the energy equivalent to the mass of an electron is 0.51 MeV, the creation of two electrons requires 1.02 MeV. Consequently, photons with energy less than 1.02 MeV do not interact by pair production. This energy requirement makes pair production irrelevant to conventional radiographic imaging. During pair production, energy in excess of 1.02 MeV is released as kinetic energy of the two electrons:

$$h\nu\ (\text{MeV}) = 1.02 + (E_k)_{e-} + (E_k)_{e+}$$

Although the nucleus recoils slightly during pair production, the small amount of energy transferred to the recoiling nucleus may usually be neglected. Pair production is depicted in Margin Figure 4-20.

Occasionally, pair production occurs near an electron rather than near a nucleus. For 10-MeV photons in soft tissue, for example, about 10% of all pair production interactions occur in the vicinity of an electron. An interaction near an electron is termed *triplet production* because the interacting electron receives energy from the photon and is ejected from the atom. Three ionizing particles, two negative electrons and one positive electron, are released during triplet production. To conserve momentum, the threshold energy for triplet production must be 2.04 MeV. The ratio of triplet to pair production increases with the energy of incident photons and decreases as the atomic number of the medium is increased.

The mass attenuation coefficient k_m for pair production varies almost linearly with the atomic number of the attenuating medium. The coefficient increases slowly with energy of the incident photons. In soft tissue, pair production accounts for only a small fraction of the interactions of x and γ rays with energy between 1.02 and 10 MeV. Positive electrons released during pair production produce annihilation radiation identical to that produced by positrons released from radioactive nuclei.



MARGIN FIGURE 4-20
Pair production interaction of a high-energy photon near a nucleus. Annihilation photons are produced when the positron and an electron annihilate each other.

Example 4-14

A 5-MeV photon near a nucleus interacts by pair production. Residual energy is shared equally between the negative and positive electron. What are the kinetic energies of these particles?

$$h\nu\ (\text{MeV}) = 1.02 + (E_k)_{e-} + (E_k)_{e+}$$

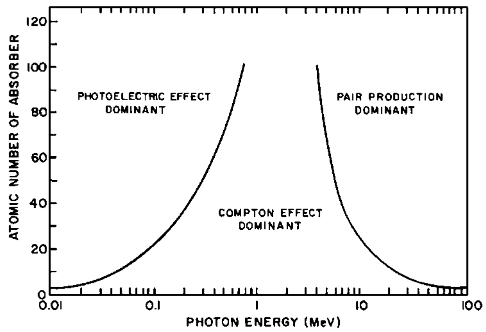
$$(E_k)_{e-} = (E_k)_{e+} = \frac{(h\nu - 1.02)\text{MeV}}{2}$$

$$= \frac{(5.00 - 1.02)\ \text{MeV}}{2}$$

$$= (E_k)_{e+} = 1.99\ \text{MeV}$$

Described in Margin Figure 4-21 are the relative importances of photoelectric, Compton, and pair production interactions in different media. In muscle or water ($Z_{\text{eff}} = 7.4$), the probabilities of photoelectric interaction and Compton scattering are equal at a photon energy of 35 keV. However, equal energies are not deposited in tissue at 35 keV by each of these modes of interaction, since all of the photon energy is deposited during a photoelectric interaction, whereas only part of the photon energy is deposited during a Compton interaction. Equal deposition of energy in tissue by photoelectric and Compton interactions occurs for 60-keV photons rather than 35-keV photons.

A summary of the variables that influence the linear attenuation coefficients for photoelectric, Compton, and pair production interactions is given in Table 4-2.



MARGIN FIGURE 4-21
Relative importance of the three principal interactions of x and γ rays. The lines represent energy/atomic number combinations for which the two interactions on either side of the line are equally probable.

TABLE 4-2 Variables that Influence the Principal Modes of Interaction of X and γ Rays

	Dependence of Linear Attenuation Coefficient on			
Mode of Interaction	*Photon Energy $h\nu$*	*Atomic Number Z*	*Electron Density ρ_e*	*Physical Density ρ*
Photoelectric	$\frac{1}{(h\nu)^3}$	Z^3	—	ρ
Compton	$\frac{1}{h\nu}$	—	ρ_e	ρ
Pair production	$h\nu$ (>1.02 MeV)	Z	—	ρ

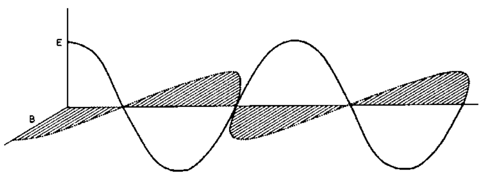
NONIONIZING RADIATION

As mentioned previously, radiation with an energy less than 13.6 eV is classified as nonionizing radiation. Two types of nonionizing radiation, electromagnetic waves and mechanical vibrations, are of interest. Electromagnetic waves are introduced here, and mechanical vibrations, or sound waves, are discussed in Chapter 19. Because no ionization is produced by the radiation, quantities such as specific ionization, linear energy transfer, and *W*-quantity do not apply. However, the concepts of attenuation and absorption are applicable.

Electromagnetic Radiation

Electromagnetic radiation consists of oscillating electric and magnetic fields. An electromagnetic wave requires no medium for propagation; that is, it can travel in a vacuum as well as through matter. In the simplified diagram in Margin Figure 4-22, the wavelength of an electromagnetic wave is depicted as the distance between adjacent crests of the oscillating fields. The wave is moving from left to right in the diagram. The constant speed c of electromagnetic radiation in a vacuum is the product of the frequency ν and the wavelength λ of the electromagnetic wave.

$$c = \lambda\nu$$



MARGIN FIGURE 4-22
Simplified diagram of an electromagnetic wave.

Often it is convenient to assign wavelike properties to electromagnetic rays. At other times it is useful to regard these radiations as discrete bundles of energy termed photons or quanta. The two interpretations of electromagnetic radiation are united by the equation

$$E = h\nu \tag{4-36}$$

where E represents the energy of a photon and ν represents the frequency of the electromagnetic wave. The symbol h represents Planck's constant, 6.62×10^{-34} J-sec.

The frequency ν is

$$\nu = c/\lambda$$

and the photon energy may be written as

$$E = hc/\lambda \tag{4-37}$$

The energy in keV possessed by a photon of wavelength λ in nanometers may be computed with Eq. (4-38):

$$E = 1.24/\lambda \tag{4-38}$$

Electromagnetic waves ranging in energy from a few nanoelectron volts up to giga-electron volts make up the electromagnetic spectrum (Table 4-3).

TABLE 4-3 The Electromagnetic Spectrum

Designation	*Frequency [hertz]*	*Wavelength [m]*	*Energy [eV]*
Gamma Rays	1.0×10^{18}–1.0×10^{27}	3.0×10^{-10}–3.0×10^{-19}	4.1×10^{3}–4.1×10^{12}
X-rays	1.0×10^{15}–1.0×10^{25}	3.0×10^{-7}–3.0×10^{-17}	4.1–4.1×10^{10}
Ultraviolet light	7.0×10^{14}–2.4×10^{16}	4.3×10^{-7}–1.2×10^{-8}	2.9–99
Visible light	4.0×10^{14}–7.0×10^{14}	7.5×10^{-7}–4.3×10^{-7}	1.6–2.9
Infra red light	1.0×10^{11}–4.0×10^{14}	3.0×10^{-3}–7.5×10^{-7}	4.1×10^{-4}–1.6
Microwave Radar and Communications	1.0×10^{9}–1.0×10^{12}	3.0×10^{-1}–3.0×10^{-4}	4.1×10^{-6}–4.1×10^{-3}
Television broadcast	5.4×10^{7}–8.0×10^{8}	5.6–0.38	2.2×10^{-7}–3.3×10^{-6}
FM radio broadcast	8.8×10^{7}–1.1×10^{8}	3.4–2.8	3.6×10^{-7}–4.5×10^{-7}
AM radio broadcast	5.4×10^{5}–1.7×10^{6}	5.6×10^{2}–1.8×10^{2}	2.2×10^{-9}–6.6×10^{-9}
Electric power	10–1×10^{3}	3.0×10^{7}–3.0×10^{5}	4.1×10^{-14}–4.1×10^{-12}

Source: Federal Communications Commission: Title 47, Code of Federal Regulations 2.106

Example 4-15

Calculate the energy of a γ ray with a wavelength of 0.001 nm.

$$E = \frac{hc}{\lambda}$$

$$E\ (\text{keV}) = \frac{(6.62 \times 10^{-34}\ \text{J-sec})(3 \times 10^{8}\ \text{m/sec})}{(\lambda\text{nm})(10^{-9}\ \text{m/nm})(1.6 \times 10^{-16}\ \text{J/keV})}$$

$$= \frac{1.24}{\lambda\text{nm}}$$

$$E = \frac{1.24}{0.001\ \text{nm}}$$

$$= 1240\ \text{keV}$$

Example 4-16

Calculate the energy of a radio wave with a frequency of 100 MHz (FM broadcast band), where 100 MHz equals $100 \times 10^6\ \text{sec}^{-1}$

The wavelength may be determined from

$$c = \lambda\nu$$

$$\lambda = c/\nu$$

$$\lambda = \frac{(3 \times 10^{8}\ \text{m/sec})(10^{9}\ \text{nm/m})}{100 \times 10^{6}\ \text{sec}^{-1}}$$

$$\lambda = 3 \times 10^{9}\ \text{nm}$$

The energy is then

$$E = \frac{1.24}{\text{nm}}$$

$$= \frac{1.24}{3 \times 10^{9}\ \text{nm}}$$

$$= 4 \times 10^{-10}\ \text{keV}$$

The wavelength of a typical radio wave is many orders of magnitude greater than the wavelength of the γ ray from the previous example. The energy of the radio wave is correspondingly smaller than that of the γ ray.

The interactions of electromagnetic waves vary greatly from one end of the electromagnetic spectrum to the other. Some of the medical uses of parts of the electromagnetic spectrum are outlined below.

Ultraviolet Light

Ultraviolet light is usually characterized as nonionizing, although it can ionize some lighter elements such as hydrogen. Ultraviolet light is used to sterilize medical instruments, destroy cells, produce cosmetic tannings, and treat certain dermatologic conditions.

Visible Light

Because it is the part of the electromagnetic spectrum to which the retina is most sensitive, visible light is used constantly by observers in medical imaging.

Infrared

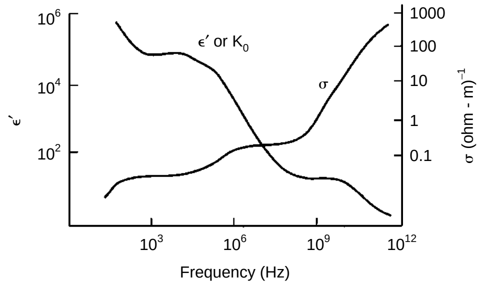
Infrared is the energy released as heat by materials near room temperature. Infrared-sensitive devices can record the "heat signature" of the surface of the body,[3] and they have been explored for thermographic examinations to detect breast cancer and to identify a variety of neuromuscular conditions. This exploration has not yielded clinically-reliable data to date.

Radar

Electromagnetic waves in this energy range are seldom detected as a pattern of "photons." Rather, they are detected by receiver antennas in which electrons are set into motion by the passage of an electromagnetic field. These electrons constitute an electrical current that can be processed to obtain information about the source of the electromagnetic wave.

AM and FM Broadcast and Television

These electromagnetic waves have resonance absorption properties with nuclei. They may be used as probes for certain nuclei that have magnetic properties in the imaging technique known as magnetic resonance imaging (see Chapter 23).



MARGIN FIGURE 4-23
Variation of dielectric constant K_0 and conductivity σ of high water content biological materials as a function of the frequency of nonionizing electromagnetic radiation. (From *Electromagnetic Fields in Biological Media.* HEW Publication No. FDA-78-8068.)

INTERACTIONS OF NONIONIZING ELECTROMAGNETIC RADIATION

For those parts of the electromagnetic spectrum with energy less than x and γ rays, mechanisms of interaction involve a direct interplay between the electromagnetic field of the wave and molecules of the target material. The energy of the electromagnetic field, and hence the energy of the wave, is diminished when electrons or other charge carriers are set into motion in the target. Energy that electrons absorb directly from the wave is referred to as conduction loss, while energy that produces molecular rotation is referred to as dielectric loss. These properties of a material are described by its conductivity σ and dielectric constant k_e. The degree to which a material is "lossy" to electromagnetic waves (that is, the material absorbs energy and is therefore heated) is a complex function of the frequency of the waves.[4]

Some generalizations about low-energy electromagnetic radiation are possible. For a given material, for example, the increase in conductivity with frequency tends to limit penetration of the radiation. Compared with lower-frequency electromagnetic waves, waves of higher frequency tend to be attenuated more severely in biological materials. Conductivity and dielectric constants are shown in Margin Figure 4-23 for typical "high-water content" biologic tissues as a function of frequency.

The rate at which energy is deposited in a material by nonionizing radiation is described by the specific absorption rate (SAR) of the material, usually stated in units of watts per kilogram. Calculation of the SAR involves consideration of the geometry of the target material as well as energy absorption parameters including conductivity, dielectric constant, and energy loss mechanisms. Specific absorption rates for typical magnetic resonance imaging pulse sequences are given in Chapter 25.

Scattering of Visible Light

Visible light, a nonionizing radiation, may interact with electrons of the atoms and molecules of a target as a group. Coherent scattering results in a change of direction of the incident photons, but essentially no change in energy. The size of the target molecules determines the characteristics of the interaction.

When light from the sun interacts with water molecules in a cloud, a type of coherent scattering called Thomson scattering occurs. Thomson scattering is equally probable for all wavelengths of visible light, and all parts of the visible spectrum are scattered in all directions. The scattered light emanating from a cloud consists of many wavelengths and is perceived as white light.[5]

When visible light from the sun encounters gas molecules of a size commonly present in air, another type of scattering called Rayleigh scattering occurs. Rayleigh scattering is an example of incoherent scattering in which the wavelength (and therefore the energy) of the radiation is changed by the interaction. Rayleigh scattering is much more likely for the shorter-wavelength components of visible light (i.e., the blue end of the spectrum).[6] Therefore, blue components of light are scattered at large angles from molecules in the sky to observers on the ground, and the sky appears blue. At sunset, when light rays reach the earth through relatively straight paths, unscattered rays excluding blue are seen. Sunsets appear red because of the absence of Rayleigh scattering of the longer-wavelength end of the electromagnetic spectrum.

PROBLEMS

*4-1. Electrons with kinetic energy of 1.0 MeV have a specific ionization in air of about 6000 IP/m. What is the LET of these electrons in air?

*4-2. Alpha particles with 2.0 MeV have an LET in air of 0.175 keV/μm. What is the specific ionization of these particles in air?

4-3. The tenth-value layer is the thickness of a slab of matter necessary to attenuate a beam of x or γ rays to one-tenth the intensity with no attenuator present. Assuming good geometry and monoenergetic photons, show that the tenth-value layer equals $2.30/\mu$, where μ is the total linear attenuation coefficient.

*4-4. The mass attenuation coefficient of copper is 0.0589 cm^2/g for 1.0-MeV photons. The number of 1.0-MeV photons in a narrow beam is reduced to what fraction by a slab of copper 1 cm thick? The density of copper is 8.9 g/cm^3.

*4-5. Copper has a density of 8.9 g/cm^3 and a gram-atomic mass of 63.56. The total atomic attenuation coefficient of copper is 8.8×10^{-24} cm^2/atom for 500-keV photons. What thickness (in centimeters) of copper is required to attenuate 500-keV photons to half of the original number?

4-6. Assume that the exponent μx in the equation $I = I_0 e^{-\mu x}$ is equal to or less than 0.1. Show that, with an error less than 1%, the number of photons transmitted is $I_0(1 - \mu x)$ and the number attenuated is $I_0 \mu$. (*Hint*: Expand the term $e^{-\mu x}$ into a series.)

*4-7. K- and L-shell binding energies for cesium are 28 keV and 5 keV, respectively. What are the kinetic energies of photoelectrons released from the K and L shells as 40-keV photons interact in cesium?

4-8. The binding energies of electrons in different shells of an element may be determined by measuring the transmission of a monoenergetic beam of photons through a thin foil of the element as the energy of the beam is varied. Explain why this method works.

*4-9. Compute the energy of a photon scattered at 45 degrees during a Compton interaction, if the energy of the incident photon is 150 keV. What is the kinetic energy of the Compton electron? Is the energy of the scattered photon increased or decreased if the photon scattering angle is increased to more than 45 degrees?

*4-10. A γ ray of 2.75 MeV from ^{24}Na undergoes pair production in a lead shield. The negative and positive electrons possess equal kinetic energy. What is this kinetic energy?

*4-11. Prove that, regardless of the energy of the incident photon, a photon scattered at an angle greater than 60 degrees during a Compton interaction cannot undergo pair production.

*For problems marked with an asterisk, answers are provided on p. 491.

SUMMARY

- Charged particles, such as electrons and alpha particles, are said to be ***directly*** ionizing.
- Uncharged particles, such as photons and neutrons, are said to be ***indirectly*** ionizing.
- LET = (SI) (W).
- Range = $\frac{E}{\text{LET}}$.
- Electrons may interact with other electrons or with nuclei of atoms.
- Inelastic scattering of electrons with a nucleus, bremsstrahlung, is used to produce x rays.

- Neutrons interact primarily by "billiard ball" or "knock on" collisions with nuclei.
- Attenuation of monoenergetic x or γ rays from a narrow beam is exponential.
- Interactions of x rays include coherent, photoelectric, Compton, and pair production.
- Liner attenuation coefficient is the fractional rate of removal of photons from a beam per unit path length.
- Examples of electromagnetic radiation in order of increasing wavelength (decreasing frequency and energy) are x and γ rays, ultraviolet, visible, infrared, and radio waves.

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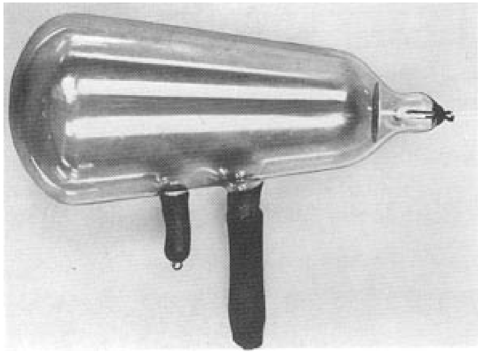
CHAPTER

5

PRODUCTION OF X RAYS

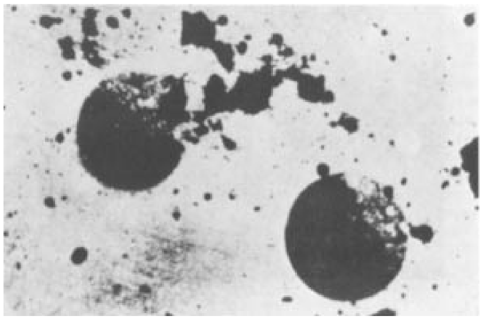
Medical Imaging Physics, Fourth Edition, by William R. Hendee and E. Russell Ritenour
ISBN: 0-471-38226-4

X rays were discovered on November 8, 1895 by Wilhelm C. Röntgen, a physicist at the University of Würzburg in Germany.[1] He named his discovery "x rays" because "x" stands for an unknown quantity. For this work, Röntgen received the first Nobel Prize in Physics in 1901.



MARGIN FIGURE 5-1
Crookes tube, an early example of a cathode ray tube.

Röntgen was not the first to acquire an x-ray photograph. In 1890 Alexander Goodspeed of the University of Pennsylvania, with the photographer William Jennings, accidentally exposed some photographic plates to x rays. They were able to explain the images on the developed plates only after Röntgen announced his discovery of x rays.



MARGIN FIGURE 5-2
An x-ray picture obtained accidentally in February 1890 by Arthur Goodspeed of the University of Pennsylvania. The significance of this picture, acquired more than 5 years before Röntgen's discovery, was not recognized by Professor Goodspeed.

Heating a filament to release electrons is called thermionic emission or the Edison effect.

Coolidge's contributions to x-ray science, in addition to the heated filament, included the focusing cup, imbedded x-ray target, and various anode cooling devices.

OBJECTIVES

By studying this chapter, the reader should be able to:

- Identify each component of an x-ray tube and explain its function.
- Describe single- and three-phase voltage, and various modes of voltage rectification.
- Explain the shape of the x-ray spectrum, and identify factors that influence it.
- Discuss concepts of x-ray production, including the focal spot, line-focus principle, space charge, power deposition and emission, and x-ray beam hardening.
- Define and apply x-ray tube rating limits and charts.

INTRODUCTION

To produce medical images with x rays, a source is required that:

1. Produces enough x rays in a short time
2. Allows the user to vary the x-ray energy
3. Provides x rays in a reproducible fashion
4. Meets standards of safety and economy of operation

Currently, the only practical sources of x rays are radioactive isotopes, nuclear reactions such as fission and fusion, and particle accelerators. Only special-purpose particle accelerators known as x-ray tubes meet all the requirements mentioned above. In x-ray tubes, bremsstrahlung and characteristic x rays are produced as high-speed electrons interact in a target. While the physical design of x-ray tubes has been altered significantly over a century, the basic principles of operation have not changed.

Early x-ray studies were performed with a cathode ray tube in which electrons liberated from residual gas atoms in the tube were accelerated toward a positive electrode (anode). These electrons produced x rays as they interacted with components of the tube. The cathode ray tube was an unreliable and inefficient method of producing x rays. In 1913, Coolidge[2] improved the x-ray tube by heating a wire filament with an electric current to release electrons. The liberated electrons were repelled by the negative charge of the filament (cathode) and accelerated toward a positive target (the anode). X rays were produced as the electrons struck the target. The Coolidge tube was the prototype for "hot cathode" x-ray tubes in wide use today.

CONVENTIONAL X-RAY TUBES

Figure 5-1 shows the main components of a modern x-ray tube. A heated filament releases electrons that are accelerated across a high voltage onto a target. The stream of accelerated electrons is referred to as the *tube current*. X rays are produced as the electrons interact in the target. The x rays emerge from the target in all directions but are restricted by collimators to form a useful beam of x rays. A vacuum is maintained inside the glass envelope of the x-ray tube to prevent the electrons from interacting with gas molecules.

ELECTRON SOURCE

A metal with a high melting point is required for the filament of an x-ray tube. Tungsten filaments (melting point of tungsten 3370°C) are used in most x-ray tubes. A current of a few amperes heats the filament, and electrons are liberated at a rate that increases with the filament current. The filament is mounted within a negatively charged focusing cup. Collectively, these elements are termed the *cathode assembly*.

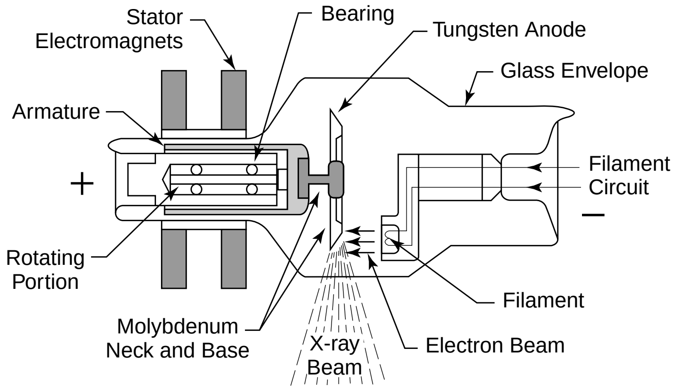


FIGURE 5-1
Simplified x-ray tube with a rotating anode and a heated filament.

The focal spot is the volume of target within which electrons are absorbed and x rays are produced. For radiographs of highest clarity, electrons should be absorbed within a small focal spot. To achieve a small focal spot, the electrons should be emitted from a small or “fine” filament. Radiographic clarity is often reduced by voluntary or involuntary motion of the patient. This effect can be decreased by using x-ray exposures of high intensity and short duration. However, these high-intensity exposures may require an electron emission rate that exceeds the capacity of a small filament. Consequently many x-ray tubes have two filaments. The smaller, fine filament is used when radiographs with high detail are desired and short, high-intensity exposures are not necessary. If high-intensity exposures are needed to limit the blurring effects of motion, the larger, coarse filament is used. The cathode assembly of a dual-focus x-ray tube is illustrated in Margin Figure 5-4.

TUBE VOLTAGE AND VOLTAGE WAVEFORMS

The intensity and energy distribution of x rays emerging from an x-ray tube are influenced by the potential difference (voltage) between the filament and target of the tube. The source of electrical power for radiographic equipment is usually alternating current (ac). This type of electricity is by far the most common form available for general use, because it can be transmitted with little energy loss through power lines that span large distances. Figure 5-2 shows a graph of voltage and current in an ac power line. X-ray tubes are designed to operate at a single polarity, with a positive

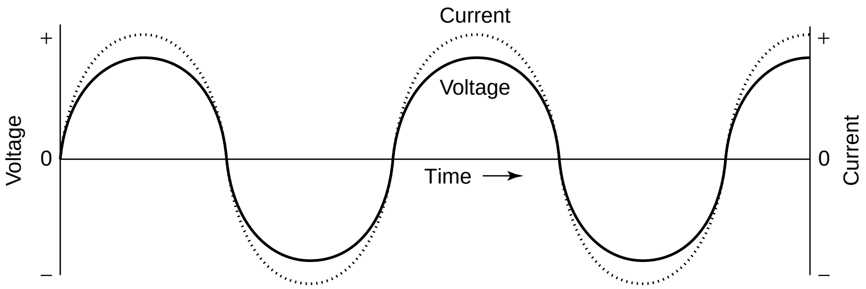
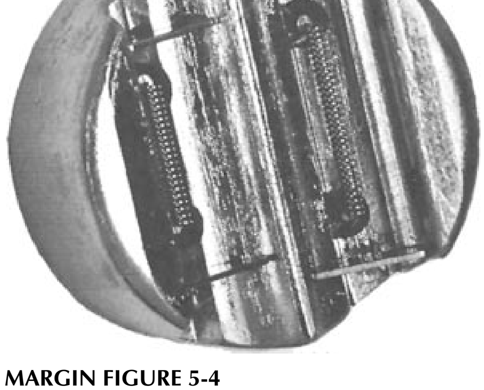


FIGURE 5-2
Voltage and current in an ac power line. Both voltage and current change from positive to negative over time. The relationship between voltage and current (i.e., the relative strength and the times at which they reach their peaks) depends upon a complex quantity called reactance. The positive and negative on the voltage scale refer to polarity, while the positive and negative on the current scale refer to the direction of flow of electrons that constitute an electric current.

Vaporized tungsten from both the filament and anode deposits on the glass envelope of the x-ray tube, giving older tubes a mirrored appearance.

An x-ray tube with two filaments is called a *dual-focus* tube.

MARGIN FIGURE 5-3
A dual-focus x-ray tube with a rotating anode.



MARGIN FIGURE 5-4
Cathode assembly of a dual-focus x-ray tube. The small filament provides a smaller focal spot and a radiograph with greater detail, provided that the patient does not move. The larger filament is used for high-intensity exposures of short duration.

A positive electrode is termed an *anode* because negative ions (anions) are attracted to it. A negative electrode is referred to as a cathode because positive ions (cations) are attracted to it.

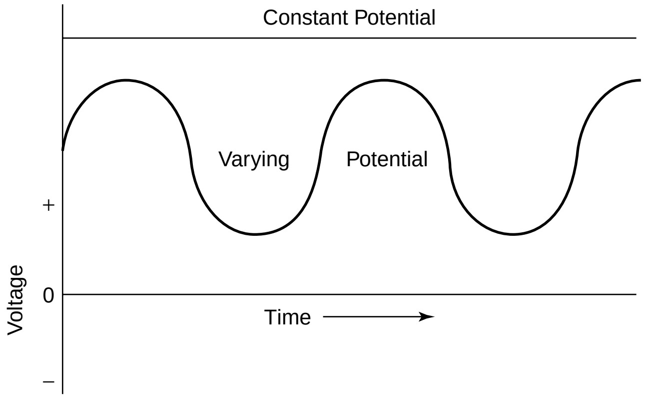


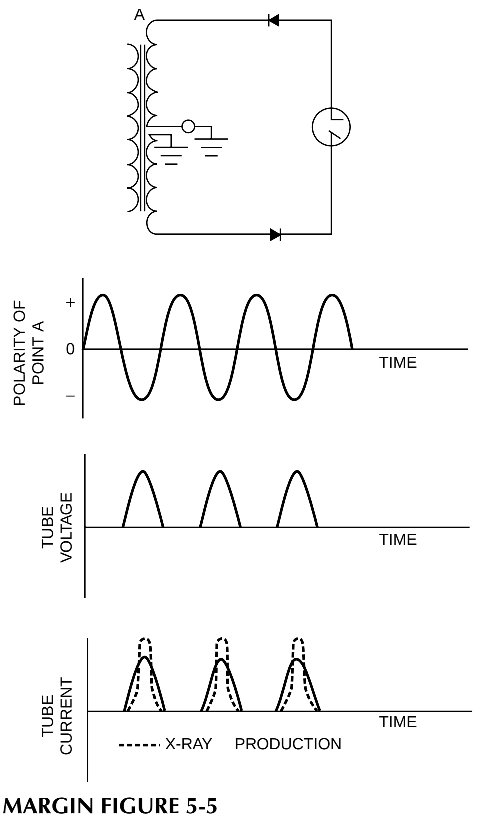
FIGURE 5-3
Voltage in a dc power line. The term *direct current* means that the voltage (and current not shown here) never reverse (change from positive to negative), although they may vary in intensity. X-ray tubes are most efficient when operated at constant potential.

The term *alternating* means that the voltage reverses polarity at some frequency (in the United States, 120 reversals per second (60 Hz) alternating current is standard).

target (anode) and a negative filament (cathode). X-ray production is most efficient (more x rays are produced per unit time) if the potential of the target is always positive and if the voltage between the filament and target is kept at its maximum value. In most x-ray equipment, ac is converted to direct current (dc), and the voltage between filament and target is kept at or near its maximum value (Figure 5-3). The conversion of ac to dc is called *rectification*.

One of the simplest ways to operate an x-ray tube is to use ac power and rely upon the x-ray tube to permit electrons to flow only from the cathode to the anode. The configuration of the filament (a thin wire) is ideal for producing the heat necessary to release electrons when current flows through it. Under normal circumstances the target (a flat disk) is not an efficient source of electrons. When the polarity is reversed (i.e., the filament is positive and the target is negative), current cannot flow in the x-ray tube, because there is no source of electrons. In this condition, the x-ray tube "self-rectifies" the ac power, and the process is referred to as *self-rectification*. At high tube currents, however, the heat generated in the target can be great enough to release electrons from the target surface. In this case, electrons flow across the x-ray tube when the target is negative and the filament is positive. This reverse flow of electrons can destroy the x-ray tube.

A rectified voltage waveform can also be attained by use of circuit components called diodes. Diodes are devices that, like x-ray tubes, allow current to flow in only one direction. A simple circuit containing diodes that produces the same waveform as *self-rectification* is shown in the margin. Rectification in which polarity reversal across the x-ray tube is eliminated is called *half-wave rectification*.



MARGIN FIGURE 5-5
A circuit for half-wave rectification (**top**), with resulting tube voltage, tube current, and efficiency for production of x rays. Rectifiers indicate the direction of conventional current flow, which is opposite to the actual flow of electrons.

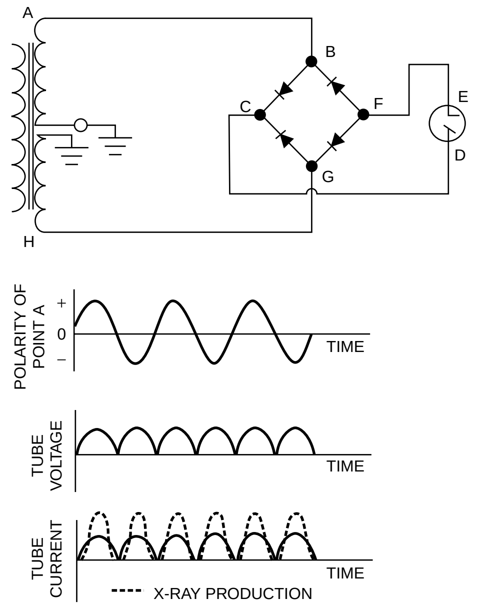
A half-wave rectifier converts ac to a dc waveform with 1 pulse per cycle. X-ray production could be made more efficient if the negative half-cycle of the voltage waveform could be used. A more complex circuit called a full-wave rectifier utilizes both half-cycles. In both the positive and negative phases of the voltage waveform, the voltage is impressed across the x-ray tube with the filament (or cathode) at a negative potential and the target (or anode) at a positive potential. This method of rectifying the ac waveform is referred to as *full-wave rectification*. In full-wave rectification the negative pulses in the voltage waveform are in effect "flipped over" so that they can be used by the x-ray tube to produce x rays. Thus a full-wave rectifier converts an ac waveform into a dc waveform having 2 pulses per cycle.

The efficiency of x-ray production could be increased further if the voltage waveform were at high potential most of the time, rather than decreasing to zero at least twice per cycle as it does in full-wave rectification. This goal can be achieved by use of three-phase (3φ) power. Three-phase power is provided through three separate voltage lines connected to the x-ray tube.

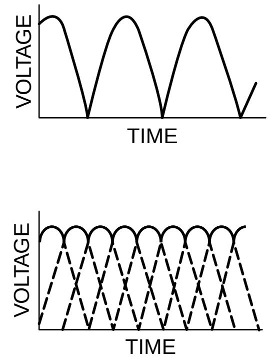
The term *phase* refers to the fact that all three voltage lines carry the same voltage waveform, but the voltage peaks at different times in each line. Each phase (line) is rectified separately so that three distinct (but overlapping) full-wave-rectified waveforms are presented to the x-ray tube. The effect of this composite waveform is to supply voltage to the x-ray tube that is always at or near maximum. In a three-phase full-wave-rectified x-ray circuit the voltage across the x-ray tube never drops to zero. With three separate phases of ac, six rectified pulses are provided during each voltage cycle.

A refinement of 3φ circuitry provides a slight phase shift for the waveform presented to the anode compared with that presented to the cathode. This refinement yields 12 pulses per cycle and provides a slight increase in the fraction of time that the x-ray tube operates near peak potential.

Modern solid-state voltage-switching devices are capable of producing "high-frequency" waveforms yielding thousands of x-ray pulses per second. These voltage waveforms are essentially constant potential and provide further improvements in the efficiency of x-ray production.



MARGIN FIGURE 5-6
A circuit for full-wave rectification (**top**), with resulting tube voltage, tube current, and efficiency for production of x rays. Electrons follow the path ABFEDCGH when end A of the secondary of the high-voltage transformer is negative. When the voltage across the secondary reverses polarity, the electron path is HGFEDCBA.



MARGIN FIGURE 5-7
Single-phase (**top**) and three-phase (**bottom**) voltages across an x-ray tube. Both voltages are full-wave rectified. The three-phase voltage is furnished by a six-pulse circuit.

RELATIONSHIP BETWEEN FILAMENT CURRENT AND TUBE CURRENT

Two electrical currents flow in an x-ray tube. The *filament current* is the flow of electrons through the filament to raise its temperature and release electrons. The second electrical current is the flow of released electrons from the filament to the anode across the x-ray tube. This current, referred to as the *tube current,* varies from a few to several hundred milliamperes.

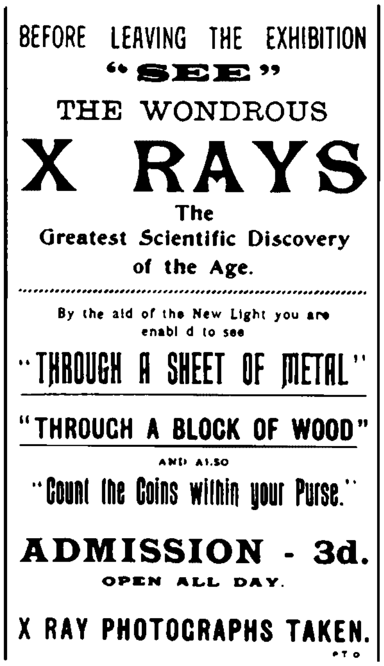
The two currents are separate but interrelated. One of the factors that relates them is the concept of "space charge." At low tube voltages, electrons are released from the filament more rapidly than they are accelerated toward the target. A cloud of electrons, termed the *space charge,* accumulates around the filament. This cloud opposes the release of additional electrons from the filament.

The curves in Figure 5-4 illustrate the influence of tube voltage and filament current upon tube current. At low filament currents, a saturation voltage is reached above which the current through the x-ray tube does not vary with increasing voltage. At the saturation voltage, tube current is limited by the rate at which electrons are released from the filament. Above the saturation voltage, tube current can be increased only by raising the filament's temperature in order to increase the rate of electron emission. In this situation, the tube current is said to be *temperature or filament-emission limited*. To obtain high tube currents and x-ray energies useful for diagnosis, high filament currents and voltages between 40 and 140 kV must be used. With high filament currents and lower tube voltages, the space charge limits the tube current, and hence the x-ray tube is said to be *space-charge limited.*

EMISSION SPECTRA

The useful beam of an x-ray tube is composed of photons with an energy distribution that depends on four factors:

- Bremsstrahlung x rays are produced with a range of energies even if electrons of a single energy bombard the target.
- X rays released as characteristic radiation have energies independent of that of the bombarding electrons so long as the energy of the bombarding electrons exceeds the threshold energy for characteristic x ray emission.
- The energy of the bombarding electrons varies with tube voltage, which fluctuates rapidly in some x-ray tubes.



MARGIN FIGURE 5-8
Poster for a public demonstration of x rays, 1896, Crystal Place Exhibition, London.

X-ray generators that yield several thousand voltage (and hence x ray) pulses per second are known as *constant potential generators*.

X-ray tubes can operate in one of two modes:

- filament-emission limited
- space-charge limited

As a rough approximation, the rate of production of x rays is proportional to $Z_{\text{target}} \times (\text{kVp})^2 \times$ mA.

Inherent filtration is also referred to as *intrinsic filtration*.

Mammography x-ray tubes often employ exit windows made of beryllium to allow low-energy x rays to escape from the tube.

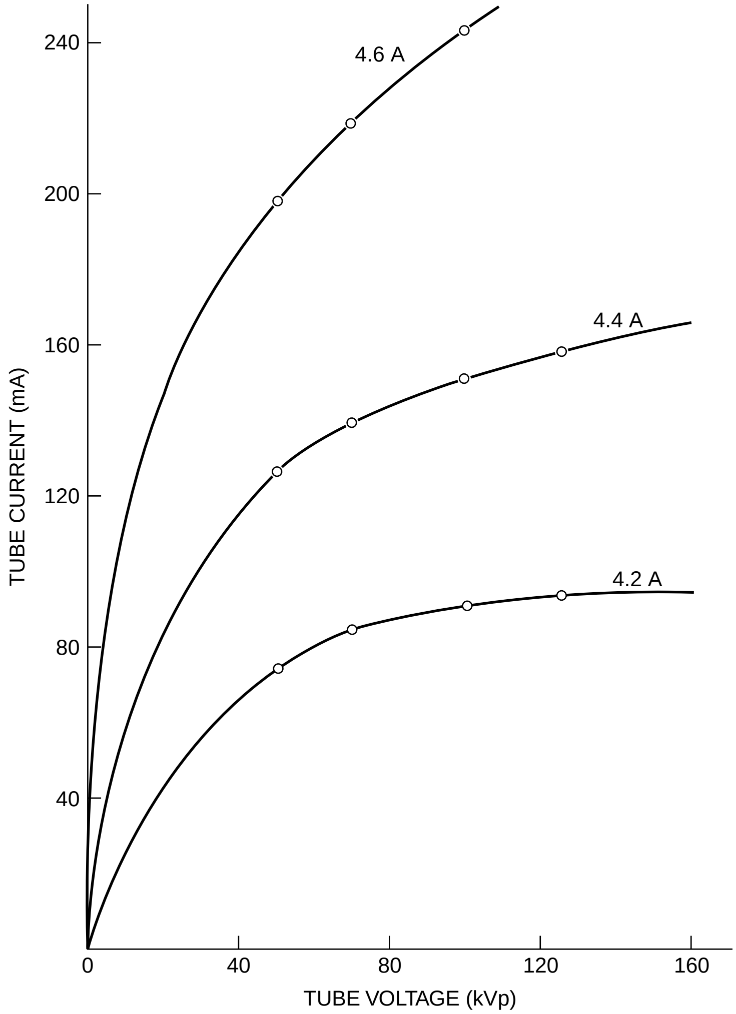


FIGURE 5-4
Influence of tube voltage and filament current upon electron flow in a Machlett Dynamax x-ray tube with a rotating anode, 1-mm apparent focal spot, and full-wave-rectified voltage.

- X rays are produced at a range of depths in the target of the x-ray tube. These x rays travel through different thicknesses of target and may lose energy through one or more interactions.

Changes in other variables such as filtration, target material, peak tube voltage, current, and exposure time all may affect the range and intensity of x-ray energies in the useful beam. The distribution of photon energies produced by a typical x-ray tube, referred to as an *emission spectrum,* is shown in Figure 5-5.

FILTRATION

An x-ray beam traverses several attenuating materials before it reaches the patient, including the glass envelope of the x-ray tube, the oil surrounding the tube, and the exit window in the tube housing. These attenuators are referred to collectively as the inherent filtration of the x-ray tube (Table 5-1). The aluminum equivalent for each component of inherent filtration is the thickness of aluminum that would reduce the exposure rate by an amount equal to that provided by the component. The inherent filtration is approximately 0.9 mm Al equivalent for the tube described in Table 5-1, with most of the inherent filtration contributed by the glass envelope. The *inherent filtration* of most x-rays tubes is about 1 mm Al.

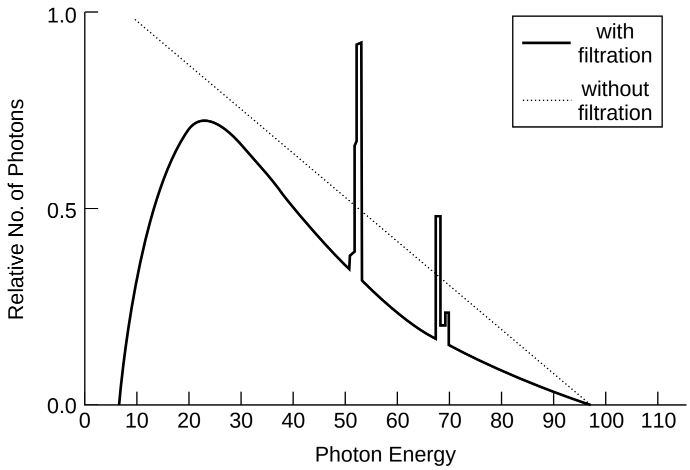


FIGURE 5-5
Emission spectrum for a tungsten target x-ray tube operated at 100 kVp. K-characteristic x-ray emission occurs for tungsten whenever the tube voltage exceeds 69 keV, the K-shell binding energy for tungsten. The *dotted line* represents the theoretical bremsstrahlung emission from a tungsten target. The *solid line* represents the spectrum after self-, inherent, and added filtration. The area under the spectrum represents the total number of x rays.

An x-ray beam of higher average energy is said to be "harder" because it is able to penetrate more dense (i.e., harder) substances such as bone. An x-ray beam of lower average energy is said to be "softer" because it can penetrate only less dense (i.e., softer) substances such as fat and muscle.

In any medium, the probability that incident x rays interact photoelectrically varies roughly as $1/E^3$, where E is the energy of the incident photons (see Chapter 4). That is, low-energy x rays are attenuated to a greater extent than those of high energy. After passing through a material, an x-ray beam has a higher average energy per photon (that is, it is "harder") even though the total number of photons in the beam has been reduced, because more low-energy photons than high-energy photons have been removed from the beam.

The inherent filtration of an x-ray tube "hardens" the x-ray beam. Additional hardening may be achieved by purposefully adding filters of various composition to the beam. The total filtration in the x-ray beam is the sum of the inherent and added filtration as shown in Table 5-1. Usually, additional hardening is desirable because the filter removes low-energy x rays that, if left in the beam, would increase the radiation dose to the patient without contributing substantially to image formation.

Emission spectra for a tungsten-target x-ray tube are shown in Figure 5-6 for various thicknesses of added aluminum filtration. The effect of the added aluminum is to decrease the total number of photons but increase the average energy of photons in the beam. These changes are reflected in a decrease in the overall height of the emission spectrum and a shift of the peak of the spectrum toward higher energy.

Equalization filters are sometimes used in chest and spine imaging to compensate for the large differences in x-ray transmission between the mediastinum and lungs.

Tube Voltage

As the energy of the electrons bombarding the target increases, the high-energy limit of the x-ray spectrum increases correspondingly. The height of the spectrum also

TABLE 5-1 Contributions to Inherent Filtration in Typical Diagnostic X-Ray Tube

Component	*Thickness (mm)*	*Aluminum-Equivalent Thickness (mm)*
Glass envelope	1.4	0.78
Insulating oil	2.36	0.07
Bakelite window	1.02	0.05

[a] Data from Trout, E. *Radiol. Technol.* 1963; **35:**161.

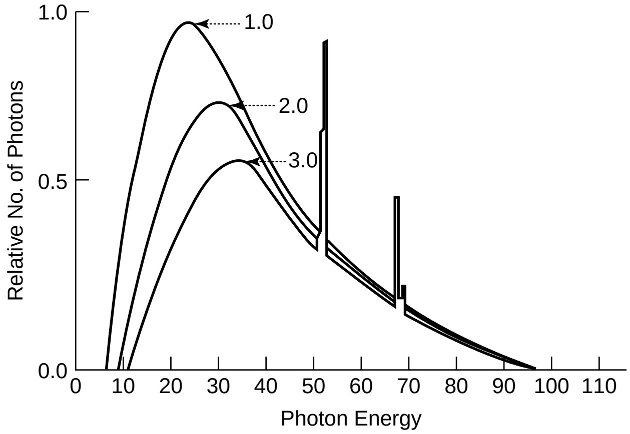
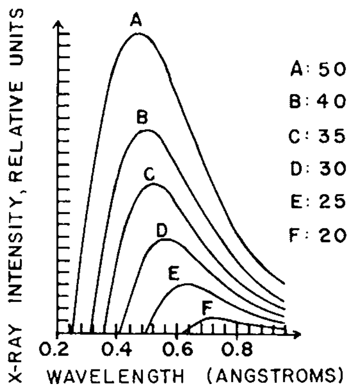


FIGURE 5-6
X-ray emission spectra for a 100-kVp tungsten target x-ray tube with total filtration values of 1.0, 2.0, and 3.0 mm aluminum. kVp and mAs are the same for the three spectra. (Computer simulation courtesy of Todd Steinberg, Colorado Springs.)



MARGIN FIGURE 5-9
Tungsten-target x-ray spectra generated at different tube voltages with a constant current through the x-ray tube. (From Ulrey, C., *Phys. Rev.* 1918; **1**:401.)

increases with increasing tube voltage because the efficiency of bremsstrahlung production increases with electron energy (see margin).

Tube Current and Time

The product of tube current in milliamperes and exposure time in seconds (mA · sec) describes the total number of electrons bombarding the target.

Example 5-1

Calculate the total number of electrons bombarding the target of an x-ray tube operated at 200 mA for 0.1 sec.

The ampere, the unit of electrical current, equals 1 coulomb/sec. The product of current and time equals the total charge in coulombs. X-ray tube current is measured in milliamperes, where 1 mA $= 10^{-3}$ amp. The charge of the electron is 1.6×10^{-19} coulombs, so

$$1 \text{ mA} \cdot \text{s} = \frac{(10^{-3}\text{coulomb/ sec})(\text{sec})}{1.6 \times 10^{-19}\text{coulomb/electron}} = 6.25 \times 10^{15}\text{electrons}$$

For 200 mA and 0.1 sec,

$$\begin{aligned}\text{No. of electrons} &= (200 \text{ mA})(0.1 \text{ sec})(6.25 \times 10^{15}\text{electrons/mA} \cdot \text{sec})\\ &= 1.25 \times 10^{17}\text{electrons}\end{aligned}$$

Other factors being equal, more x rays are produced if more electrons bombard the target of an x-ray tube. Hence the number of x rays produced is directly proportional to the product (mA · sec) of tube current in milliamperes and exposure time in seconds. Spectra from the same x-ray tube operated at different values of mA · sec are shown in Figure 5-7. The overall shape of the spectrum (specifically the upper and lower limits of energy and the position of characteristic peaks) remains unchanged. However, the height of the spectrum and the area under it increase with increasing mA · sec. These increases reflect the greater number of x rays produced at higher values of mA · sec.

Target Material

The choice of target material in an x-ray tube affects the efficiency of x-ray production and the energy at which characteristic x rays appear. If technique factors (tube voltage, milliamperage, and time) are fixed, a target material with a higher atomic

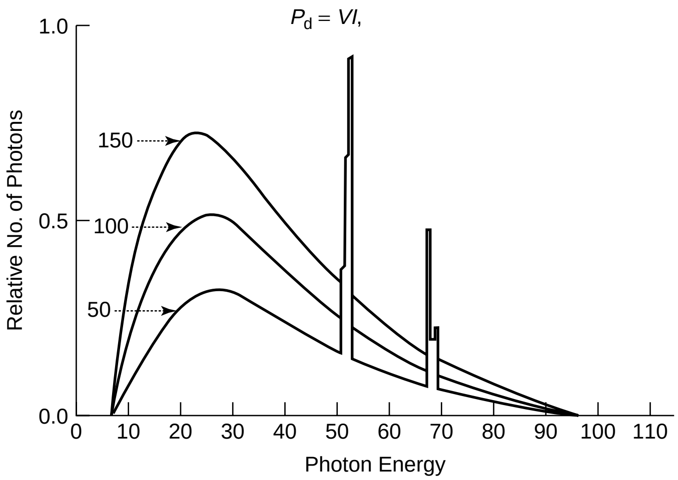


FIGURE 5-7
X-ray emission spectra for a 100-kVp tungsten target x-ray tube operated at 50, 100, and 150 mA. kVp and exposure time are the same for the three spectra.

number (Z) will produce more x rays per unit time by the process of bremsstrahlung.

The efficiency of x-ray production is the ratio of energy emerging as x radiation from the x-ray target divided by the energy deposited by electrons impinging on the target. The rate at which electrons deposit energy in a target is termed the *power deposition* P_d (in watts) and is given by

$$P_d = VI$$

In the equation $P_d = VI$, the voltage may also be expressed in kilovolts if the current is described in milliamperes.

where V is the tube voltage in volts and I is the tube current in amperes. The rate at which energy is released as x radiation,[3] termed the *radiated power* P_r, is

$$P_r = 0.9 \times 10^{-9} Z V^2 I \qquad \textbf{(5-1)}$$

where P_r is the radiated power in watts (W) and Z is the atomic number of the target. Hence, the efficiency of x-ray production is

$$\begin{aligned} \text{Efficiency} &= \frac{P_r}{P_d} = \frac{0.9 \times 10^{-9} Z V^2 I}{VI} \qquad \textbf{(5-2)} \\ &= 0.9 \times 10^{-9} Z V \end{aligned}$$

Equation (5-2) shows that the efficiency of x-ray production increases with the atomic number of the target and the voltage across the x-ray tube.

Example 5-2

In 1 sec, 6.25×10^{17} electrons (100 mA) are accelerated through a constant potential difference of 100 kV. At what rate is energy deposited in the target?

$$\begin{aligned} P &= (10^5 \text{V})(0.1 \text{A}) \\ &= 10^4 \text{W} \end{aligned}$$

X-ray production is a very inefficient process, even in targets with high atomic number. For x-ray tubes operated at conventional voltages, less than 1% of the energy deposited in the target appears as x radiation. Almost all of the energy delivered by impinging electrons is degraded to heat within the target.

Efficiency of converting electron energy into x rays as a function of tube voltage.[4]

kV	Heat (%)	X Rays (%)
60	99.5	0.5
200	99	1.0
4000	60	40

The characteristic radiation produced by a target is governed by the binding energies of the K, L, and M shells of the target atoms. Theoretically, any shell could contribute to characteristic radiation. In practice, however, transitions of electrons among shells beyond the M shell produce only low-energy x rays, ultraviolet light,

TABLE 5-2 Electron Shell Binding Energies (keV)

Shell	*Molybdenum* ($Z = 42$)	*Tungsten* ($Z = 74$)
K	20	69
L	2.9, 2.6, 2.5[a]	12, 11, 10
M	0.50, 0.41, 0.39, 0.23, 0.22	2.8, 2.6, 2.3, 1.9, 1.8

[a] Multiple binding energies exist within the L and M shells because of the range of discrete values of the quantum number *l*, discussed in Chapter 2.

and visible light. Low-energy x rays are removed by inherent filtration and do not become part of the useful beam. The characteristic peak for a particular shell occurs only when the tube voltage exceeds the binding energy of that shell. Binding energies in tungsten and molybdenum are shown in Table 5-2.

A characteristic x ray released during transition of an electron between adjacent shells is known as an α x ray. For example, a K_α x ray is one produced during transition of an electron from the L to the K shell. A β x ray is an x ray produced by an electron transition among nonadjacent shells. For example, a K_β x ray reflects a transition of an electron from the M to the K shell.

The characteristic radiation produced by an x-ray target is usually dominated by one or two peaks with specific energies slightly less than the binding energy of the K-shell electrons. The most likely transition involves an L-shell electron dropping to the K shell to fill a vacancy in that shell. This transition yields a photon of energy equal to the difference in electron binding energies of the K and L shells. A characteristic photon with an energy equal to the binding energy of the K shell alone is produced only when a free electron from outside the atom fills the vacancy. The probability of such an occurrence is vanishingly small.

During interaction of an electron with a target nucleus, a bremsstrahlung photon may emerge with energy equal to the total kinetic energy of the bombarding electron. Such a bremsstrahlung photon would have the maximum energy of all photons produced at a given tube voltage. The maximum energy of the photons depicted in Figure 5-5 therefore reflects the peak voltage applied across the x-ray tube, described in units of kilovolts peak (kVp). Photons of maximum energy $E_{\max}$ in an x-ray beam possess the maximum frequency and the minimum wavelength.

$$E_{\max} = h\nu_{\max} = \frac{hc}{\lambda_{\min}}$$

The minimum wavelength in nanometers for an x-ray beam may be computed as

$$\lambda_{\min} = \frac{hc}{E_{\max}} = \frac{hc}{\text{kVp}}$$

In the past, x-ray wavelengths were described in units of angstroms (Å), where Å $= 10^{-10}$ m. The minimum wavelength in angstroms would be $\lambda_{\min}$ (Å) $= 12.4/\text{kVp}$. The angstrom is no longer used as a measure of wavelength.

where $E_{\max}$ is expressed in keV.

$$\lambda_{\min} = \frac{(6.62 \times 10^{-34}\ \text{J-sec})(3 \times 10^{8}\ \text{m/sec})(10^{9}\ \text{nm/m})}{\text{kVp}(1.6 \times 10^{-16}\ \text{J/keV})}$$

$$\lambda_{\min} = \frac{1.24}{\text{kVp}} \tag{5-3}$$

with $\lambda_{\min}$ expressed in units of nanometers (nm).

Example 5-3

Calculate the maximum energy and minimum wavelength for an x-ray beam generated at 100 kVp.

The maximum energy (keV) numerically equals the maximum tube voltage (kVp). Because the maximum tube voltage is 100 kVp, the maximum energy of the photons is 100 keV:

Compared with visible light, x rays have much shorter wavelengths. Wavelengths of visible light range from 400 nm (blue) to 700 nm (red).

$$\lambda_{\min} = \frac{1.24}{100\ \text{kVp}}$$
$$= 0.0124\ \text{nm}$$

TUBE VACUUM

To prevent collisions between air molecules as electrons accelerate between the filament and target, x-ray tubes are evacuated to pressures less than 10^{-5} Hg. Removal of air also reduces deterioration of the hot filament by oxidation. During the manufacture of x-ray tubes, evacuation is accomplished by "outgassing" procedures that employ repeated heating cycles to remove gas occluded in components of the x-ray tube. Tubes still occasionally become "gassy," either after prolonged use or because the vacuum seal is not perfect. Filaments are destroyed rapidly in gassy tubes.

Many x-ray tubes include a "getter circuit" (active ion trap) to remove gas molecules that otherwise might accumulate in the x-ray tube over time.

ENVELOPE AND HOUSING

A vacuum-tight glass envelope (the "x-ray tube") surrounds other components required for the efficient production of x rays. The tube is mounted inside a metal housing that is grounded electrically. Oil surrounds the x-ray tube to (a) insulate the housing from the high voltage applied to the tube and (b) absorb heat radiated from the anode. Shockproof cables that deliver high voltage to the x-ray tube enter the housing through insulated openings. A bellows in the housing permits heated oil to expand when the tube is used. Often the bellows is connected to a switch that interrupts the operation of the x-ray tube if the oil reaches a temperature exceeding the heat storage capacity of the tube housing. A lead sheath inside the metal housing attenuates radiation emerging from the x-ray tube in undesired directions. A cross section of an x-ray tube and its housing is shown in Figure 5-8.

Secondary electrons can be ejected from a target bombarded by high-speed electrons. As these electrons strike the glass envelope or metallic components of the x-ray tube, they interact to produce x rays. These x rays are referred to as off-focus x rays because they are produced away from the target.

Off-focus radiation can also be reduced by using small auxiliary collimators placed near the output window of the x-ray tube.

The quality of an x-ray image is reduced by off-focus x rays. For example, off-focus radiation contributes as much as 25% of the total amount of radiation emerging from some x-ray tubes with rotating anodes.[5] The effects of off-focus radiation on images may be reduced by placing the beam collimators as close as possible to the x-ray target.

For x-ray images of highest quality, the volume of the target from which x rays emerge should be as small as possible. To reduce the "apparent size" of the focal spot,

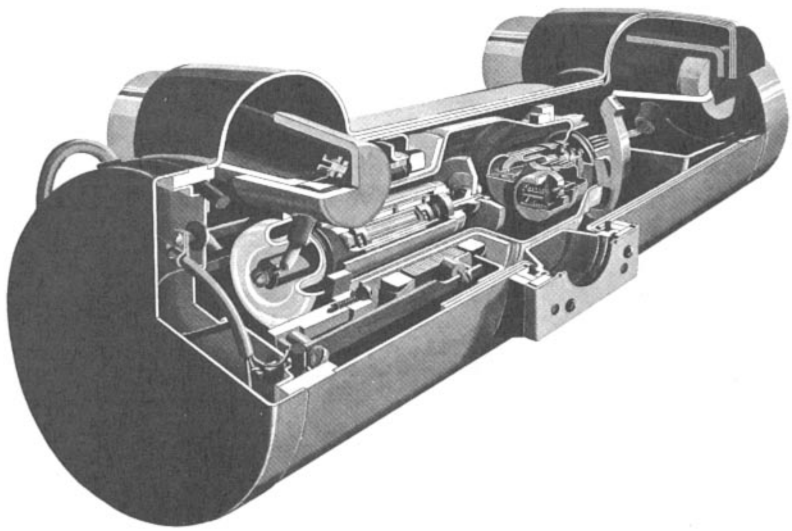


FIGURE 5-8
Cutaway of a rotating-anode x-ray tube positioned in its housing. (Courtesy of Machlett Laboratories, Inc.)